

Scale-Coupled Dynamics: Gravity and Charge from a Single Non-Commuting Scale Field

Xinyu Yang 

(Dated: August 12, 2026)

We develop an axiomatic theory in which *scale* — a dimensionless local unit ratio — rather than spacetime is primitive. From seven axioms we derive that the local scale must be directional (a scalar version is refuted by its own consequences), that two scalings along different directions must fail to commute with an antisymmetric residue that is exactly a rotation, that the resulting Lie algebra has real rank one and hence belongs to the Lorentz family, and that the number of directions is fixed by a single postulate about what counts as an observation. Curvature, the gauge connection, and the renormalization group turn out to be different readings of one object — the scale vector field and its own rate of change. The isotropic sector reproduces the Schwarzschild metric and closes quantitatively: light bending $1.7515''$ and Mercury’s perihelion advance $42.989''/\text{century}$, both to the precision quoted, with no free parameter beyond the single dimensionless unit any dimensionless theory must carry. The anisotropic sector produces point particles as unresolvable knots of the scale vector field, carrying exactly a $\mathbb{Z}$ winding charge and a $\mathbb{Z}/2$ confined charge. Non-commutativity of the scale generators produces $\hbar$ as an exact step along the scale axis and makes the entire quantum correction to curvature purely antisymmetric, proving that the radiative sector of gravitational waves is identical to general relativity at *every* post-Newtonian order — the two theories differ, if at all, only through the absence of a horizon. We confront the framework with GW250114’s area-law test, the ergoregion-instability constraint from rapidly spinning remnants, DESI’s dark-energy equation of state, the $f_0(500)$ pole width, and fourteen decades of the measured particle spectrum, and report honestly where the framework currently loses: a $2.8\text{--}4.2\sigma$ tension with DESI’s preference for an evolving equation of state, and an open question at the $f_0(500)$ pole. The argument has been verified end to end in the Lean 4 proof assistant — 650 machine-checked theorems, zero admitted gaps — reproduced as ordinary mathematics in the Supplemental Material, which this self-contained paper does not require to be read.

I. INTRODUCTION

General relativity and the Standard Model both take a four-dimensional spacetime manifold as given and place fields upon it. Each has open seams at the place where it meets the other, or meets its own foundations: a singularity where curvature diverges and the classical description simply stops; a cosmological constant whose value is measured, not explained, by either theory; a renormalization group whose running couplings are computed order by order in perturbation theory rather than obtained in closed form; a particle spectrum whose labels — charge, color, generation — are read off experiment and then built into the group-theoretic structure of the Lagrangian by hand. None of these are failures of either theory on its own terms; they are the places where a choice was made — a manifold, a gauge group, a set of fields — that the theory itself does not derive.

This paper explores a different starting point, one deliberately more austere than either: take the *scale* at a point — a dimensionless ratio of a local unit to a reference — as the only primitive, and ask what follows if that scale is directional, because the physical sources that produce it (energy-momentum) are tensorial rather than scalar, and non-commuting, because performing two scalings along different directions in the wrong order leaves a residue. Spacetime curvature, the gauge connection, particle content, and $\hbar$ are then not independently postulated; they are different questions asked of a single object,

the scale vector field, and several of the open seams listed above turn out to be forced closed by that object’s own algebra rather than left as free choices: the dimension and signature of spacetime, the particle-label alphabet, the leading term of the renormalization group, and the relation between the metric connection and the gauge connection all become theorems rather than inputs.

This is not a claim that spacetime does not exist, nor a claim that the program competes with approaches that quantize the metric directly (loop quantum gravity), posit extended objects and extra dimensions (string theory), or build spacetime from discrete causal order (causal set theory). Those programs start by choosing what is fundamental among geometry, matter, and causal structure, and then quantize or discretize it. This paper asks a prior question: whether geometry, gauge structure, and particle content are themselves derivable from something more primitive than any of the three — and answers it constructively, by exhibiting the derivation rather than arguing for its possibility in the abstract. The claim is that the four-dimensional, Lorentzian, curved structure usually assumed can instead be *derived* from a single forced algebraic fact: two symmetric operators bracket to an antisymmetric one. Everything that follows is the working-out of that one fact’s consequences, and nothing else is smuggled in along the way — a claim that Section XIV exists specifically to make checkable rather than asserted.

Every derivation summarized here has an independent, complete proof; we have verified the entire argument me-

chanically in the Lean 4 proof assistant against no axioms beyond those of standard mathematics (`propext`, `Classical.choice`, and `Quot.sound`), and the Supplemental Material reproduces every statement and proof — 650 theorems in total, spanning definitions, lemmas, and closing numerical bounds — in full mathematical detail, organized to mirror this paper section by section. This paper does not depend on that verification to be read; it is a self-contained physics argument, written so that a reader who never opens the Supplemental Material still receives every derivation’s actual content rather than a pointer to it. What the verification adds is a guarantee that cannot be obtained by proofreading alone: that every “Theorem” in what follows is exactly what it claims to be, with every hypothesis stated and no step skipped, because a proof assistant does not accept an argument it cannot check line by line.

The remainder of the paper is organized as follows. Section II states the physical picture in ordinary language before any algebra, because a statement built from commutators and a Lie-algebra classification is not obviously less abstract than the curved geometry it replaces, unless a picture is attached to it first; the picture introduces, informally, every structural result the later sections derive formally, so that the algebra which follows always has a physical question attached to it. Section III states the seven axioms precisely. Section IV derives the forced chain from the axioms to a four-dimensional Lorentzian structure carrying both a metric connection and a gauge connection — the single argument on which the rest of the paper depends. Section V states the branch point between the isotropic and anisotropic sectors, which determines whether a given region of the scale field carries curvature alone or curvature together with charge. Section VI gives the isotropic (gravitational) sector in full: the Schwarzschild solution and the two classical solar-system tests, the Newtonian limit obtained exactly rather than as a truncated series, local covariance under global dissipation, the absence of singularities and horizons, and gravitational-wave dynamics from inspiral through ringdown, including a concrete confrontation with GW250114 [1] and GW241011 [2]. Section VII gives the anisotropic (particle) sector: the two topological charges, their independence, and the resulting bound on the mass spectrum. Section VIII gives the renormalization group, obtained as a theorem about a defect density rather than a perturbative computation, and the strong-interaction scale. Section IX gives the quantum sector — $\hbar$ as an exact step along the scale axis — and the all-orders equivalence with general relativity in the gravitational-wave sector that follows from it. Section X gives dark matter, dark energy, and the scale-blindness of light. Section XI collects seven questions that only this framework’s own primitives can pose, as a test of whether it is more than a paraphrase of existing physics. Section XII confronts the framework with data, in one table. Section XIII states, observation by observation, what would kill which part of the framework. Sec-

tion XIV describes how the argument was verified and records what has been retracted, corrected, cited without proof, and assumed along the way — the standing answer to whether any of the foregoing rests on more than the seven axioms plus one external physical input. Section XV concludes.

II. THE PHYSICAL PICTURE

Sixteen statements, in the order they compose, with the algebra deferred to the sections that follow. Each previews a result proved formally later; the section reference at the end of a paragraph is where the informal claim becomes a theorem.

There is no space, only a ratio. At every point there is a number, and that number is a ratio: the local unit compared to some reference. The choice of reference is unobservable by construction, so only *differences* of this number are meaningful. Picture a map that records, everywhere, how long a meter is there — with no grid drawn on it. The grid, i.e. spacetime, is what emerges from the *inconsistency* between these numbers at nearby points, not something the numbers are written onto. This is a stronger claim than “spacetime is emergent” in the sense usually meant by that phrase: it is not that a smooth manifold approximates some discrete substrate at long wavelengths, but that the manifold structure itself — the fact that nearby points have well-defined relations at all — is read off differences in a single scalar-valued field, with no independent metrical input.

“Two places” is constructed, not primitive. If the scale pattern at two points is parallel, no observable distinguishes them — not two places that look alike, but no second place at all, since every question one could ask of one point has the identical answer at the other. A perfectly uniform region has no internal structure and no extent: distance is the accumulated *non-parallelism* between points, and because the wedge product of two vectors is antisymmetric, separation is naturally a signed area, not a length. Two consequences follow immediately, both proved in Section IV: a truly homogeneous universe would be a single point no matter how large a coordinate volume one assigns it, and the ordinary notion of distance as a positive scalar is recovered only after choosing an orientation, which this framework does not supply for free.

Scale is directional because energy-momentum is tensorial. At each point the scale is not a number but a *vector*: a fluid carries different energy density along its flow than transverse to it, and a tensorial source cannot be matched by a single scalar scale. This is not an aesthetic preference for more structure; Section VI proves the scalar version of this requirement is not too weak but too strong — it *forbids* the Schwarzschild metric outright, so a scalar scale is not merely insufficient to describe the solar system, it actively excludes the possibility of describing it. Directionality is consequently not an enrichment chosen

for generality; it is the minimum needed for the theory to be compatible with a single measured orbit.

Gravity is the ruler changing, not space bending. A change in the *magnitude* of the scale vector across position is exactly gravity; curvature is the second-order inhomogeneity of scale — not a first-order gradient, which is unobservable, but the rate at which that gradient itself changes. Nothing is bent: light follows what looks locally straight, but the ruler used to judge straightness changes along the path, so the trajectory appears curved. Tidal forces are the *second-order* change in the ruler — the first-order, uniform-dilation part is unobservable, being exactly a reference shift. Two bodies in free fall separate or approach not because a force acts between them but because the local unit of length differs infinitesimally at their two positions, and that difference is what a tidal-force meter actually measures; the entire content of the gravitational field, on this reading, is a statement about how a ruler's length varies from place to place, encoded completely in the second derivative of one scalar field per direction (Section VI).

Rotation is direction changing — and it is generated by scaling. A change in the *direction* of the scale vector across position is rotation, and here is the theory's sharpest forced step: performing two scalings along different directions in the two possible orders must differ by a rotation. This is not a chosen coupling; it is forced by the meaning of the words — a scaling is a pure stretch and hence symmetric, a rotation preserves length and hence is antisymmetric, and the bracket of two symmetric operators is necessarily antisymmetric (Theorem 2). Physically this is the Thomas–Wigner rotation [3], and the Terrell–Penrose effect [4, 5] is its visual form: a fast-moving object is *seen* as rotated rather than contracted, because a change of scale is being read as a rotation. One direction: flat. Two or more: curvature, generated by the scalings themselves, with no separate assumption about how gravity and rotation relate — they are the symmetric and antisymmetric parts of one algebra, and the bracket between them is arithmetic, not physics chosen to fit.

The gauge connection and the gravitational connection are the same object. Both are built from the same log-derivative of the scale pattern, evaluated on two different loci of the same field: where the pattern is directionally uniform (isotropic), the connection reads as curvature; where it is not (anisotropic), the same construction reads as a Yang–Mills field strength. A gauge theory and a metric theory are, in this framework, two questions asked of one connection, not two structures glued together after the fact (Section IV).

“Unifying the forces” is a question in a vocabulary this theory does not have. The standard unification program asks for one gauge group from which every force descends, which presupposes that forces are couplings and that couplings are the fundamental description; in this framework neither holds, since comparisons of local units are primitive and forces are read off them. The question this

framework can answer instead is what is expressible at all: a quantity is observable — invariant under A5's fiducial shift — exactly when it is a function of scale *differences*, not merely typically or approximately, so the scale channel expresses differences and nothing else, and an absolute scale value is consequently not expressible by any construction available here, not merely unmeasured. This is also the reason no single cross-sector test between the scale channel and the rotation channel exists: in the fully scalar reading the two channels are independent by construction, one being an abelianization and the other its kernel, so neither can constrain the other — and Section IV's directional coupling is exactly what breaks that independence once scale is given a direction, replacing “independent channels” with “one channel generating the other.”

Whether charge exists at all is itself a fact about scale. The branch between the isotropic and anisotropic loci is not fixed once and for all; it is a property of the scale pattern at a given resolution, and the same region of the substrate can sit on one side of the branch at one energy scale and the other side at a different one. This is the framework's reading of the empirical fact that interactions separate as energy falls: it is not that several independent forces happen to decouple at different scales, but that whether an order parameter, and hence a charge, is present at all is a scale-dependent question with a single underlying source (Section V).

$\hbar$ is a step along the scale axis. Not a mysterious constant: translating one step along the scale axis fails to commute with position, and the size of the mismatch is $\hbar$ — exactly, not as a first-order approximation in some formal deformation parameter that is later set to a physical value. Uncertainty is then not measurement disturbance but the mutual exclusion of scale sharpness and content richness: a perfectly sharp scale is a single point on the scale axis, and a point carries no content, so resolving any finite amount of structure requires a finite window, and the width of that window times the density of what it must resolve is bounded below (Section IX).

The entire quantum correction to geometry is a rotation, never a stretch. Once the scale algebra is made non-commutative, the correction it produces to curvature is exactly the commutator of two scale generators — and a commutator, by the same symmetric/antisymmetric split that produces gravity and rotation in the classical theory, is purely antisymmetric. The symmetric, radiative part of the geometry is consequently untouched by quantization to any order: gravitational waves are the one observable sector in which this framework and general relativity cannot differ, not merely one in which the difference is currently too small to see (Section IX).

“What exists” is a slice, not a property of the world. Changing resolution does not mean knowing more about the same particles; it means looking at a different set of them. “The list of fundamental particles” is accordingly ill-posed: there is always unresolved content above any scale, with no final layer. The size of the Standard

Model's label set measures the logarithmic range it covers, not physical depth — a theory that must remain valid over thirty orders of magnitude in energy needs a label for everything that appears anywhere in that range, whether or not it is realized at any particular scale one happens to probe.

A particle is an unresolvable knot in the scale vector field. Not a field excitation but a configuration: the vector winds around a point and cannot be combed flat. Its integer charge is the winding number of the vector around a sphere; its mass is not a property it carries but the location at which it appears on the scale axis; its rotational label is $v \wedge \nabla v$, so charge necessarily comes with spin. A second, independent label — a confinement class rather than a winding number — is available from the same construction, and the two are logically independent of each other: knowing one fixes nothing about the other (Section VII). Exactly two labels are available, not three and not one, because the scale pattern in the anisotropic sector has already been shown to collapse to a single vector (Section IV), and a single vector supports exactly a winding number and a monodromy class and nothing further.

The renormalization group is a theorem about counting, not a perturbative expansion. A coupling read as a ratio of defect densities changes with the logarithm of scale at a rate fixed directly by the same dissipation constant that drives cosmic expansion (Section VIII). No diagram is summed and no loop order is chosen; the leading term of the beta function is an algebraic consequence of what a coupling *is* in this framework; a subject usually organized around successive orders of a perturbative expansion is organized here around a single exact linear relation.

The universe is not expanding; scale is running. Galaxies are not flying apart — the ruler is shrinking, and because energy scale is the reciprocal of length scale, the energy scale is dissipating. There is no first moment, because there is no zero scale: scale is a unit and is never zero, so “the age of the universe” is not a quantity — what exists is an elapsed ratio, finite and measurable, with no origin (Section X).

Black holes are arbitrarily dark, never fully black. The horizon is where the time component of the metric vanishes, and that component is minus the square of a scale unit — never zero. No singularity, no horizon, no infinite redshift, all three the same fact: ratios are invertible. A black hole is a region of extreme but finite scale ratio; the exterior geometry is untouched, so shadow, ISCO, and the early ringdown all agree with general relativity, but nothing is causally cut off and no information is destroyed (Section VI).

A horizonless remnant should ring the way a black hole does, then echo. Because nothing in the metric is ever exactly reflecting or exactly absorbing, a gravitational-wave signal that would terminate at a horizon in general relativity instead partially reflects back and forth in this framework, producing a train of echoes evenly spaced in

time after the main ringdown, with amplitude falling geometrically from one echo to the next. This is the one place this framework's predictions are qualitatively distinctive without being quantitatively computable, since the reflectivity itself depends on physics — the interior equation of state — that is explicitly outside this framework's scope (Section VI).

The two arrows of time are one order. A long-standing puzzle elsewhere is why the direction of entropy increase coincides with the direction of cosmic expansion; the usual answer invokes a special initial condition aligning two otherwise independent facts. Here they are not two facts. Three results, proved independently and never invoking one another — that time is the direction along which scale drifts (Section IV), that an H -theorem for coarse-grained content holds using only finite resolution, with no scale dependence anywhere in its own proof, and that the entropy of a region of scale ratio R is $\rho \ln R$ (Section X) — combine into the following.

Theorem 1 (The two arrows are one order). *For $\rho > 0$ and $R_1, R_2 > 0$,*

$$\rho \ln R_1 \leq \rho \ln R_2 \iff R_1 \leq R_2,$$

and likewise with strict inequalities.

Proof. $\rho \ln$ is a strictly increasing function of R for fixed $\rho > 0$, since $\ln$ is strictly increasing and multiplication by a positive constant preserves order. $\square$

Ranking regions by entropy and ranking them by scale ratio are *literally* the same order relation, not two arrows that happen to align. The one modeling choice this rests on is reading entropy as $\rho \ln R$, the count of resolvable thresholds within a region of scale ratio R ; that reading is motivated by, but not forced by, the threshold-density construction of Section VIII, so the identity is exact given the reading and the reading itself is a choice, stated as such.

Summary. A number (ratio) becomes a vector because its source is tensorial; the vector's magnitude changing is gravity, its direction changing is rotation; the two are coupled by the meaning of the words involved; the commutator of that coupling gives $\hbar$ and the quantum correction; where the vector cannot be combed flat is a particle; the vector running overall is cosmology. All physics here is a different reading of the same vector field. What this picture does *not* buy is stated equally plainly: no coupling constants, no mass values, no generation structure, no scattering amplitudes. It answers “what,” not “how much”; the quantitative part closes fully only in the gravitational sector, and the particle sector delivers label structure and bounds, not numbers.

III. AXIOMS

The substrate is a commutative differential ring over n directions; an earlier version of this development posited

three distinct carrier structures for different sectors of the theory, later shown to reduce to one. Seven axioms are used throughout.

Axiom 1 (A1 — substrate). *There is a commutative ring A with n commuting derivations $d_1, \dots, d_n : A \rightarrow A$, each additive and satisfying the Leibniz rule.*

Axiom 2 (A2 — scale field). *A distinguished unit $s \in A^\times$, the local scale.*

Axiom 3 (A3 — scale-energy duality). *An energy unit $\varepsilon \in A^\times$ with $\varepsilon \cdot s = 1$.*

Axiom 4 (A4' — directional measurement). *Scale is assigned per direction: $s : \{1, \dots, n\} \rightarrow A^\times$, not a single scalar. (An earlier, scalar version, A4, is refuted in Section VI: it forbids the Schwarzschild metric outright.)*

Axiom 5 (A5 — fiducial covariance). *Physical content is invariant under $s \mapsto u \cdot s$ for any global unit u : only ratios of scale are observable.*

Axiom 6 (A6 — openness / dissipation). *The universe is not a closed system: its total energy scale ε dissipates monotonically, $\dot{\varepsilon} = -\lambda\varepsilon$ for a constant rate $\lambda \neq 0$.*

Axiom 7 (A7 — observability). *A commitment about what counts as a single observation, formalized as a dimension-matching condition between the rotation sector the coupling generates and the direction sector A4' labels; it fixes the number of independent directions rather than leaving it a free integer.*

A7 is deliberately an axiom and not an informal judgment: it is a commitment about what an observation is rather than a claim about the world, and Section XIII returns to the consequence that it cannot be falsified by any single experiment.

A6 is stated narrowly on purpose, exactly as formalized: a statement about one scalar quantity, the total energy scale, not about the substrate's full multi-directional drift. Several places below — the signature argument of Section IV, the local-covariance argument of Section VI, the no-global-field-equation argument of Section XI — read A6 as an instance of a broader informal principle, that no global potential accounts for the drift in *any* direction, not only the energy-scale one. That broader reading is not itself derived from the axiom as stated; it is an extrapolation used consistently throughout, registered as such in Section XIV rather than left to blend into the axiom's own wording.

One further ingredient, the scale response law

$$e^{2\sigma} R = \kappa \rho, \quad (\text{A6}')$$

relating the curvature scalar R built from the scale field to a matter density ρ , is used in the gravity sector. **It is not one of the seven axioms.** It is an external physical input — the one place general relativity's own field equation is assumed rather than derived — and it is registered as such in Section XIV.

IV. THE FORCED CHAIN TO A FOUR-DIMENSIONAL LORENTZIAN GAUGE STRUCTURE

This section is the spine of the paper: a single chain of implications, each step forced rather than chosen, from A1–A7 to a four-dimensional Lorentzian structure carrying both a metric and a gauge connection.

The scale/rotation split is not a choice. Given A1–A4', any endomorphism of the tangent data at a point decomposes canonically into a symmetric part (a scaling) and an antisymmetric part (a rotation); this is a fact about the algebra of endomorphisms of an inner-product space, not a modeling decision, since every such endomorphism is the sum of its symmetric and antisymmetric parts and that decomposition is unique. An earlier stage of this development reached the same split by a different route, an abelianization of an abstract comparison group applicable only to the scalar case; that construction is superseded by, and is a special case of, the endomorphism decomposition used from here on, and the two are not independently combined into a stronger joint claim.

Time is the direction along which scale drifts. Given A6 (dissipation), exactly one direction carries a definite, non-cancelling sign of drift; the signature is forced to a codimension-one split, $(1, n-1)$. A theory in which every direction could drift in either sign with no distinguished one would admit no consistent notion of “before” and “after”; A6 rules that out. Labeling the distinguished direction *timelike* rather than merely *distinguished* is a convention consistent with the split, not a further consequence A6 forces on its own — a distinction registered explicitly rather than smoothed over (Section XIV).

The bookkeeping form is $\tau(xy)$ for a trace τ alone, not an independent postulate: any bilinear form on the ring compatible with both the ring product and the derivations must take this shape, because a trace is (up to scale) the unique linear functional invariant under conjugation, and conjugation invariance is what compatibility with the derivations demands.

Curvature is the commutator of transports. Let ∇_i denote parallel transport of a tensorial quantity along direction i , built directly from the scale pattern rather than postulated independently. Then the curvature of the connection is

$$F_{ij} := [\nabla_i, \nabla_j], \quad (1)$$

literally the failure of two transports to commute, rather than an independently defined tensor whose vanishing is imposed or assumed. Two consequences follow directly from this definition rather than from extra input: first, commuting derivations ($d_i d_j = d_j d_i$, part of A1) does not imply a flat connection — F_{ij} can be nonzero even when the underlying derivations commute, because curvature lives in the connection, not in the chart; second, the

same commutator F_{ij} is what is computed in the gravitational sector as the Riemann curvature and what is computed in the anisotropic sector as the Yang–Mills field strength (Section VII) — **gravity and gauge structure are the same object, read under two different questions about which sector’s scale pattern is being transported.** Neither is assumed; both are the same commutator. This identity extends one step further, to the quantum sector of Section IX: the commutator $[D_i, D_j]$ that gives curvature here is the identical construction that gives $\hbar$ there, evaluated on a different family of transports — phase-space translations in one case, frame transports in the other. Curvature and $\hbar$ are consequently not merely analogous; they are one algebraic construction read on two different transport families, a stronger and more specific claim than any analogy between gravity and quantum mechanics usually offered.

A further split, orthogonal to the isotropic/anisotropic branch of Section V, separates transport itself into a scale part and a frame-rotation part. The scale part, being a gradient by A5, is automatically integrable — mixed partial derivatives commute — and contributes nothing whatsoever to the curvature F_{ij} ; all of it is carried by the frame-rotation part. This is the structural reason this framework avoids the defect that killed Weyl’s original 1918 unified theory [6], where an analogous non-Riemannian connection produced a length that depended on the path taken between two points (a “second clock effect” ruled out by observation): here the scale transports integrably by construction, so no such path-dependence can arise, while curvature — gravity — lives entirely in the non-commuting frame-rotation sector. Gravity is consequently not, even informally, “the scale field bending something”; it is the failure of frame rotations specifically to commute, with the scale itself never contributing a discrepancy of its own.

Directions are the defining representation, not the algebra itself. Once the algebra generated by the scalings and rotations is named explicitly, its dimension as a Lie algebra ($\dim \mathfrak{so}(3,1) = 6$ for the eventual case $n = 4$) exceeds the number of directions (4), so directions form the vector representation on which the algebra acts, not the algebra’s own coordinates — a distinction an earlier version of this development conflated.

Theorem 2 (Coupling is forced). *Let Σ_i be the scaling generator along direction i . Then $[\Sigma_i, \Sigma_j]$ is nonzero for $i \neq j$ and is purely antisymmetric, i.e. a rotation generator.*

Proof. A scaling is symmetric with respect to the pairing induced by the trace form (it preserves direction and only rescales magnitude, so it is self-adjoint); a rotation is antisymmetric (it preserves length, so its generator satisfies $R^T = -R$). For symmetric A, B ,

$$(AB - BA)^T = B^T A^T - A^T B^T = BA - AB = -(AB - BA),$$

so the commutator of two symmetric operators is antisymmetric. That the bracket is additionally *nonzero* for

$i \neq j$ — that the coupling is not merely well-defined but genuinely active — is witnessed concretely rather than argued in general: an explicit boost pair in three spatial dimensions exhibits a nonvanishing bracket, and no general nondegeneracy argument beyond that concrete witness is claimed. $\square$

No coupling constant is chosen here; the coupling exists purely because of what the words “scaling” and “rotation” mean algebraically. The generated rotation is not merely known to exist and be antisymmetric; its concrete form is fixed once the two scaling directions are: for scalings along direction vectors v, w , the rotation their bracket generates is exactly the antisymmetric bilinear pairing $(v \wedge w)_{ij} = v_i w_j - v_j w_i$, the wedge of the two directions. This makes the coupling’s output a genuine two-form — antisymmetric, bilinear, and vanishing exactly when the two scaling directions are parallel — rather than an abstract nonzero element asserted only to exist; a wound configuration in the anisotropic sector, whose direction genuinely varies rather than staying parallel to itself, consequently carries this rotational label automatically, while a homogeneous pattern — one direction wedged with itself — carries none, whatever magnitude it has.

Real rank one, hence the Lorentz family. The Lie algebra generated by one vector’s worth of scalings together with the rotations they force has real rank one: the maximal abelian subalgebra of diagonalizable elements is one-dimensional, since a second independent scaling direction would, by Theorem 2, generate a rotation mixing it with the first, destroying simultaneous diagonalizability. Citing, without reproving, the classification of real-rank-one real simple Lie algebras — $\mathfrak{so}(n,1)$, $\mathfrak{su}(n,1)$, $\mathfrak{sp}(n,1)$, $\mathfrak{f}_{4(-20)}$ — this places the algebra in the Lorentz family $\mathfrak{so}(n,1)$; the other branches require additional structure (a complex, quaternionic, or octonionic module structure on the directions) absent from a real scale-vector construction, leaving $\mathfrak{so}(n,1)$ as the surviving case. The step from “one drift functional” to “one-dimensional maximal abelian subalgebra” above is a physical identification of what counts as a single independent scaling direction, not a formal deduction from A1–A7 alone — stated here rather than left implicit, since it is exactly the kind of step where a numerical coincidence can hide a choice (Section XIV).

$n = k + 1$, and $k = 3$ from A7. The number of directions n is one (time) plus the number of independent spatial directions k . k is not free once A7’s dimension-matching condition is imposed: requiring the number of rotations the coupling generates to match the number of directions A4’ labels gives $k = 3$. This is not the only matching condition one could write down — comparing the dimension of the full algebra to the dimension of its own defining representation instead gives $k = 2$, and the two conditions provably disagree — so A7 is doing real work: it is a specific commitment about *which* two quantities must be equal, not an arithmetic inevitability. The

framework’s own argument for preferring the first condition is that the labels available are the directions A4’ actually assigns scales to, and the coupling’s output must match *those*, not the abstract size of the algebra that generates it — but this is an argument for a modeling choice, not a proof that no other choice is coherent, and it is why A7 is registered as a postulate about what counts as an observation rather than presented as forced by pure counting.

The pattern is one vector; more axes are unavailable. An apparent dilemma between a “uniaxial” and a “biaxial” reading of an earlier theorem is resolved once the relevant vanishing identity, $v \wedge v = 0$, is recognized as a statement about *homogeneity* — a vector wedged with itself is always zero, regardless of how many axes exist — rather than a constraint on the number of axes. Once that misreading is removed, the scale pattern is shown to collapse to exactly one vector, and additional independent axes are not merely disfavored but structurally unavailable given the coupling of Theorem 2.

The chain therefore delivers, from A1–A7 with no further input: a Lorentzian signature, four dimensions, a metric and a gauge connection that are the same underlying commutator, and a coupling between scaling and rotation forced by algebra rather than fit to data.

V. THE BRANCH POINT

The scale pattern at a point is either *isotropic* — the same in every direction — or *anisotropic*, and whether it is one or the other is a scale-dependent fact, not a fixed property of the point:

isotropic	anisotropic
no order parameter	order parameter present
no charge	charges present
no multiplet	multiplets present
no rotation-sector charge	curvature/rotation active

Both columns have one source: isotropy is a single algebraic condition on the pattern (uniform direction across the pattern’s components), and charge, multiplet structure, and rotation-sector activity all switch on or off together with it, not independently. The formal branch condition itself is a static property of a given pattern, not a function of a separate energy-scale variable; the further, motivating reading — that the same physical region can present as isotropic at one probing resolution and anisotropic at another, giving this framework’s account of interactions separating as energy falls — is an interpretation connecting this branch to the scale-flow structure of Section VIII, not itself a theorem proved from the branch condition alone, and is stated here as such. Section VI treats the isotropic sector; Section VII the anisotropic one.

An apparent contradiction is worth forestalling: Section VI computes curvature in a sector called “isotropic.”

“Isotropic” here means uniform *direction* of the scale vector, not uniform *magnitude* — a radially varying but directionally uniform scale pattern has no charge and no rotation-sector label, and still has curvature, because curvature is generated by magnitude inhomogeneity (Section II), which is an axis independent of direction uniformity.

A second consequence of the branch structure fixes what happens *inside* a multiplet, *given* an assignment of one scaling operator to each direction determined by that direction’s own scale value — a hypothesis satisfied automatically whenever such an assignment exists, though real rank one (Section IV) only guarantees one vector and one scaling overall, not a free per-direction assignment, so the statement below is conditional rather than a further automatic consequence of the chain. Under that hypothesis, two directions assigned to the same multiplet cannot rotate into each other without the underlying scale pattern itself changing first: a scaling generator bracketed with *itself* vanishes identically, and two directions sharing a multiplet by definition share a scale value, hence share a scaling generator, hence generate no internal rotation between them — a binary fact about equal versus unequal scale values, not a count that decreases gradually with multiplet size. Any change in which directions belong to which multiplet accordingly requires a change in the underlying scale pattern itself: multiplet structure cannot rotate on its own, it can only be re-partitioned by the pattern that defines it changing. What is unconditional, established directly in Section VII without this hypothesis, is that the real-rank-one pattern’s own rotation is carried entirely by its variation, exactly as Section II states.

A. What “sector” means, and what regresses under which conditions

The word *sector* has been doing two different jobs in this paper, and naming them separately removes what has, in early readings, been the single most confusing feature of this framework. One question is the branch just proved: whether the scale pattern at a point is isotropic or anisotropic. A second, independent question, taken up in Section IX, is whether the underlying algebra is read as commutative (A1’s own baseline) or as its non-commutative, $\hbar$ -deformed extension. Neither question is a different choice of axioms: A1–A7 hold everywhere, unconditionally, and never change. The first question asks about a *value* the axioms’ own objects happen to take at a point — whether a given scale pattern is directionally uniform, a fact about that pattern, not about the rules governing it. The second asks whether one is working in the commutative ring A1 already postulates or in its canonical non-commutative enlargement, a strictly larger structure that contains the commutative ring exactly, as the $\hbar \rightarrow 0$ limit, rather than replacing it. “Sector,” in other words, names a place in a fixed structure, not a

different rule set adopted for convenience — which is also why quantum vocabulary appearing inside the gravity section, or charge vocabulary appearing with no $\hbar$ in sight, is not a category error: the two questions are logically independent, and ordinary quantum mechanics, for instance, needs only the second, at any point, isotropic or not. Table I lists what each answer, or combination of answers, currently delivers.

Condition	Recovers	Status
Isotropic, commutative	Newtonian gravity, then Schwarzschild GR	exact; identical to GR to all post-Newtonian orders in the symmetric part
Any point, non-commutative	Ordinary quantum mechanics $\hbar$ (CCR, uncertainty)	exact at first order in $\hbar$
Isotropic, non-commutative	A quantum-gravitational correction to curvature	structural only: purely antisymmetric, coupled to spin, magnitude uncomputed
Anisotropic, commutative	Particle labels, confinement, Yang–Mills-shaped curvature	label structure and bounds, not masses or couplings
Anisotropic, non-commutative	—	not reached: no scattering amplitudes, no quantized gauge theory here

TABLE I. The two independent questions “sector” asks, and what each of their combinations currently delivers.

The blank last row is worth one further word rather than none: the combination it names — charge-carrying, quantized, genuinely dynamical — is where a reader familiar with the Standard Model would expect quantum electrodynamics or chromodynamics to sit, with particle creation, annihilation, and scattering built from the same $\hbar$ -deformation used in the isotropic sector but applied to an anisotropic pattern instead. Nothing in this paper constructs that combination; the deformation of Section IX has only been carried through for the isotropic case, and extending it to a pattern that already carries charge is an open problem this framework does not currently solve, not a result being withheld.

That two independent questions asked of one field can look, from outside, like a family of separate theories — mechanics, general relativity, quantum mechanics, a nascent quantum gravity — each with its own domain and its own vocabulary, is not a coincidence this framework is positioned to explain; it is compatible with a broader and older observation in philosophy of science, that physical knowledge often arrives as a patchwork of locally valid models rather than a single seamless law [7]. Nothing here accounts for why that patchwork historically took the shape it did, or resolves any tension between its pieces. What can be said, and no more, is that

within this one construction the boundaries between the pieces are answers to two explicit questions about a single field rather than boundaries between different kinds of stuff.

VI. THE ISOTROPIC SECTOR: GRAVITY

A. The Schwarzschild solution and the two classical tests

One honest cost of the isotropic sector is worth stating before its results: over a commutative ring the general transport-commutator construction of Section IV gives *no* curvature at all, since the commutator that defines it vanishes identically on commuting scalars — this is exactly why the isotropic geometry needs its own, separately built curvature tensor rather than inheriting one from the chain’s general machinery, and it is a cost confined entirely to this classical sector, not a limitation on the coupling itself, the quantum sector, or the anisotropic sector, all of which keep their transports genuinely non-commutative. With the scale pattern isotropic in direction, a metric is built directly from the scale field: writing the local scale along the time and radial directions as s_t, s_r , the line element takes the diagonal form $ds^2 = -s_t^2 dt^2 + s_r^2 dr^2 + r^2 d\Omega^2$. A deformation tensor built directly from second derivatives of the scale field, in the specific combination that the Riemann tensor is forced to equal for a conformally flat metric, gives the Ricci tensor component by component for this static, spherically symmetric ansatz; the vacuum condition (Ricci flat away from a source) becomes the stationarity of a particular sum of scale-derivative terms rather than an independently imposed differential equation. Imposing asymptotic flatness — $s_t, s_r \rightarrow 1$ at spatial infinity, the only boundary condition consistent with A5 (no preferred finite reference scale) — fixes the single remaining integration constant by itself, with no further input: the stationarity condition only pins down the *derivative* of $\sigma_t + \sigma_r$, and asymptotic flatness alone supplies the boundary value that makes the constant zero. A6’ plays no role in this particular step; it enters later, in the Newtonian limit and in fixing the overall normalization of the source term. Reciprocity

$$s_t \cdot s_r = 1 \quad (2)$$

follows as a theorem, not an ansatz: the Schwarzschild form is recovered exactly, not approximately, as the unique solution consistent with the axioms and the one external input A6’.

From $s_t s_r = 1$, the PPN parameter is exactly $\gamma = 1$, and the two classical solar-system tests follow with no

further assumption:

$$\delta\phi_{\odot} = 1.7515'' \quad (\text{measured } 1.7509 \pm 0.0002'' \text{ [8]}) \quad (3)$$

$$\Delta\varpi_{\text{Mercury}} = 42.989''/\text{cent.} \quad (\text{measured } 42.98 \pm 0.04'' \text{ [4]}) \quad (4)$$

Both bounds are certified to the precision quoted. Two internally excluded alternatives are worth stating explicitly because they show the framework can fail its own tests: computing deflection or precession from the isotropic sector's magnitude change *alone*, discarding the directional structure of Section IV, gives $0.8758''$ and $14.33''$ respectively — both wrong by a clean integer factor (2 and 3) relative to the measured values. Each is excluded by the same underlying apparatus but through a distinct step: the deflection case is excluded by the theorem that full isotropy together with reciprocity forces flat space outright, while the precession case is excluded by the same $\gamma = 1$ derivation used for deflection, evaluated at the isotropic sector's $\gamma = 0$ instead. Neither exclusion is merely a comparison with data.

B. The Newtonian limit and gravitational redshift

Linearizing $s_t = e^{\sigma}$ about the asymptotic value $\sigma = 0$, the scale response law A6' reduces, to leading order in σ , to a Poisson-type equation for σ sourced by the same density ρ that appears in A6', with the reciprocal constant fixed by the same asymptotic-flatness argument used above. Since $g_{tt} = -s_t^2 = -e^{2\sigma} \approx -(1 + 2\sigma)$, a weak, static source gives the standard Newtonian potential to leading order, recovering Newtonian gravity as the linear regime of the same reciprocity relation that gives $1.7515''$ at higher order — not a separately posited limit. Gravitational redshift between two static observers at different σ follows directly from the ratio of their local scale units, $s_t(\sigma_2)/s_t(\sigma_1)$, and evaluated for a laboratory height difference this reproduces the Pound–Rebka–Snider result [9, 10] to the precision of Table II — an arithmetic check of the A3 reading of scale as a clock rate, not an independent prediction, since it uses the same reciprocity relation already fixed above.

The linear regime above is not an approximation whose error one hopes is small; it is realized *exactly* by extending the substrate ring to the dual numbers $A[\epsilon]/(\epsilon^2)$, a square-zero infinitesimal extension, and setting $\sigma = \epsilon\varphi$ for an ordinary function φ . Every derivation carries over componentwise to this extended ring, and the quadratic invariant $|\nabla\sigma|^2$ vanishes *identically* rather than merely being small, since $\sigma_i^2 = \epsilon^2(\partial_i\varphi)^2 = 0$ by $\epsilon^2 = 0$. Poisson's equation $2(n-1)\Delta\Phi = \kappa\rho$ therefore holds with no term discarded and no linearization error, for the entirely algebraic reason that there is nothing to discard: the “weak-field limit” is not a truncation of the exact theory but an exact statement about a different, infinitesimally-deformed ring, related to the finite theory only by which numbers one is willing to admit as the value of σ .

C. Local covariance survives global dissipation

A6 makes the theory globally non-conservative: there is no functional whose gradient reproduces the total drift of the scale field. This does not threaten local Lorentz covariance, because the two facts live at different scopes. The coupling of Theorem 2 and the Lorentz algebra of Section IV are pointwise, algebraic facts, holding identically at every point regardless of how the scale field as a whole evolves; A6 is a global, integrated statement about the *sum* of the drift across all points. A local Lorentz frame is constructed at each point from the coupling alone and is exactly as covariant there as in a theory with a globally conserved potential; openness only says that no single potential accounts for the drift *everywhere at once*, which is compatible with, and does not weaken, covariance at any one point.

D. No singularity, no horizon

Because $g_{tt} = -s_t^2$ with s_t a unit — nonzero everywhere by A2 — g_{tt} never vanishes: there is no horizon and no curvature singularity. The exterior geometry away from the would-be horizon is unchanged from Schwarzschild, so shadow shape, ISCO location, and the early ringdown spectrum are unaffected; the difference between this framework and general relativity is concentrated entirely in the boundary condition at small radius, where general relativity imposes absorption (a horizon) and this framework cannot, since nothing in it is ever exactly zero.

The absence of a singularity is not merely the absence of a place where s_t hits zero; it is structural at the level of what curvature is built from. The Riemann tensor of Section IV is a fixed linear function of a deformation tensor built entirely from the *first and second derivatives* of σ — the scale value s itself never appears in the curvature formula, only its rate of change. Two scale fields with identical derivatives at every point therefore have identical curvature even if their values differ arbitrarily, and a region where the scale becomes extreme is on that account alone no more curved than a region where it does not: what general relativity would read as an approach to a singularity, this framework reads as a region where σ is large in magnitude but changing no faster than elsewhere, which is exactly what asymptotic flatness combined with reciprocity $s_t s_r = 1$ guarantees, since a large σ in one direction is bought by an equally large and opposite σ in the reciprocal direction, without either derivative diverging.

A scale ratio, being built from units, is itself a unit — never zero, never infinite, but with no upper bound on its magnitude either. There is consequently no minimum resolvable length and, on the matter side of the same statement, no maximum density (Section XI): for any proposed shortest scale a finer one is always available, so there is no final effective description this framework's

own structure ever forces a halt at, only successive replacement at each threshold crossed.

E. Two polarizations, from two different sectors, and no breathing mode

A transverse gravitational wave in k spatial dimensions can, a priori, carry perturbations from two independent sources at each point on the propagation plane: a change in the local scale magnitude (the scaling sector) and a change in the local scale direction (the rotation sector), matching the split of Section IV. In the two transverse directions available for $k = 3$, the scaling sector contributes $\dim(\text{transverse}) - 1 = 1$ independent mode — not 2, because a uniform rescaling of both transverse directions together is exactly a fiducial shift, and A5 declares fiducial shifts unobservable, removing that mode from the physical spectrum entirely rather than merely making it small. The rotation sector contributes $\dim \Lambda^2(\text{transverse}) = 1$ independent mode, the wedge of the two transverse directions. The total is

$$1_{\text{scale}} + 1_{\text{rotation}} = 2, \quad (5)$$

matching the two tensor polarizations of gravitational-wave observation, but arrived at as a sum of contributions from two structurally distinct sectors rather than as a single count. This gives a discriminator sharper than the count alone: the framework has **no scalar (breathing) mode at any amplitude**, since that mode is precisely the uniform rescaling A5 removes, whereas scalar–tensor theories of gravity generically predict one at some coupling-dependent amplitude. A breathing-mode detection would not merely disfavor this framework; A5 gives it nowhere to appear at all, so the correct statement is not “predicted small” but “predicted absent.” This also bears, incidentally, on the ultralight-boson exclusion drawn from GW241011’s rapid spin [2]: that analysis rules out a scalar of Compton wavelength in the super-radiant band, and this framework predicts no such field to begin with, since its label structure (threshold, $\mathbb{Z}/2$, $\mathbb{Z}$) has no further slot for one and the scalar polarization it would carry is exactly what A5 removes — the non-detection needs no explanation here, where a scalar-tensor theory owes one.

F. Gravitational waves: what radiates, and what does not

Conservation is not an additional postulate laid alongside a field equation for the scale field; it is the condition for such an equation to be writable at all. If curvature is proportional to a source — the shape of any field equation — the Bianchi identity forces that source’s cyclic covariant divergence to vanish identically, with no further input, since Bianchi is itself a consequence of the associativity of transport (the Jacobi identity for

the transport bracket). The contrapositive is the sharp form: a source whose cyclic divergence fails to vanish cannot be the curvature source of *any* connection, so this framework does not merely prefer conserved sources over non-conserved ones — it has no expression for a non-conserved source at all. Conservation, read this way from the Bianchi identity as a statement about the scale field’s own algebra, kills the monopole and dipole moments of radiation specifically. This is established purely algebraically: divergences form the additive subgroup generated by images of the non-drift derivations, and because A1’s derivations commute, the drift direction descends consistently to the quotient by that subgroup — no measure, no Stokes’ theorem, and no decay condition at infinity is used; what survives is one purely algebraic condition, that this subgroup of divergences is proper rather than the whole ring. Dipole radiation is accordingly identically zero, consistent with the double-pulsar bound $|\dot{P}_b^{\text{excess}}/\dot{P}_b| < 1.3 \times 10^{-4}$; the quadrupole leads, giving the standard inspiral chirp.

The one constant κ supplied by A6’ appears exactly once, not twice: it fixes both the Newtonian-limit Poisson normalization and the quadrupole radiation coefficient, so fixing it from the static, weak-field limit leaves the radiated power with no remaining freedom — a binary pulsar’s orbital decay is consequently a genuine test rather than a fit to a second free constant. For a circular binary with component masses m_1, m_2 and separation r , the orbital energy $E = -Gm_1m_2/2r$ and the quadrupole radiated power $P = \frac{32}{5}G^4(m_1m_2)^2(m_1+m_2)/(c^5r^5)$ together force the orbit to shrink, $dr/dt < 0$, at a closed-form rate obtained by nothing more than the chain rule applied to $dE/dt = -P$. The gravitational-wave frequency evolution depends on the two masses only through the chirp mass $\mathcal{M} = (m_1m_2)^{3/5}/(m_1+m_2)^{1/5}$, via $df/dt \propto \mathcal{M}^{5/3}f^{11/3}$ — the same combination and the same two exponents, 5/3 and 11/3, that a gravitational-wave detector template actually fits. The chirp mass combination and its two exponents are read off directly from the same quadrupole formula used for the binary-pulsar prediction; the further step from the time-domain separation rate to this frequency-domain form uses Kepler’s third law in its standard, uncontested form, which is cited rather than re-derived here — the exponents are not an independent input chosen to match observation, but the derivation is not carried through in full from first principles either. Applied to PSR B1913+16 (Hulse–Taylor) [11, 12], the predicted-to-observed ratio of orbital decay is 0.9983, agreeing to two parts in a thousand and bounding any dipole contamination below 0.2%; applied to the double pulsar PSR J0737–3039 [13], sixteen years of timing bound the same dipole fraction below 1.3×10^{-4} , an order of magnitude tighter, with every improvement in timing precision tightening a prediction that has no parameter left to retune.

Theorem 3 (All-order equivalence with general relativity). *The symmetric (radiative) part of the deformation tensor equals its classical, general-relativistic counterpart*

at every *post-Newtonian* order.

This is an exact statement, not a leading-order approximation, and its origin is given in Section IX: the entire quantum correction to curvature is purely antisymmetric, so the symmetric sector carries none of it, at any order. There is accordingly no order at which the two theories' gravitational-wave predictions could differ — not merely an order too small to currently measure. The gravitational-wave sector can discriminate this framework from general relativity only through the ringdown boundary condition, the presence or absence of a horizon, never through the inspiral or the post-Newtonian expansion itself.

G. The horizon-absence signature: echoes and the ergoregion bound

Absence of a horizon permits, but does not predict the amplitude of, late echoes. Writing $A_{\min}$ for the field's reflectivity at the would-be horizon (a quantity this framework does not compute, Section XII A) and τ for the light-travel time to that surface, the round-trip delay is $\tau_{\text{echo}} = \tau \ln(1/A_{\min})$ — logarithmic, not linear, in the redshift $1/A_{\min}$: ten orders of magnitude of additional redshift lengthen the delay by only a factor of order ten, not 10^{10} , which is why an echo search looks for a comb of arrivals at millisecond spacing rather than at a Planck time. The n -th echo arrives at $n\tau_{\text{echo}}$, evenly spaced, with amplitude falling as $A_0 R^{2n}$ for reflectivity R after n round trips — a null search excluding an amplitude above some fraction b of the ringdown directly bounds $R^2 \leq b$.

Two independent observational handles exist, and they scale with R differently. Echo searches place a direct bound on R that is polynomial in the excluded amplitude fraction, as above. Ergoregion (superradiant) instability of a rapidly spinning, long-lived compact remnant places an *exponential* bound on $\mathcal{R}$, because a horizonless spinning object whose reflectivity is not exponentially close to unity is unstable on an observationally short timescale — the growth rate of the instability, not merely the detector's sensitivity to a faint echo, sets the bound, so a longer-lived remnant tightens the constraint exponentially rather than linearly. GW241011's [2] primary spin, $\chi_1 = 0.64 > 0.55$ at the reported 90% lower limit, is in the rapid regime, and the remnant's evident long-term stability makes this the tighter of the two constraints currently available — and, honestly stated, a real and previously unregistered pressure on this framework: because $\mathcal{R}$ is not computable here (Section XII A), the framework cannot currently say whether it survives this constraint, only that the constraint applies and tightens with every additional rapidly-spinning long-lived remnant observed. The comparison between GW250114's [1] ringdown and its inspiral-predicted remnant is itself a genuine confirmation, independent of the reflectivity question: the inferred final mass ($62.7 M_\odot$, from components 33.6 and $32.2 M_\odot$

radiating $3.1 M_\odot$, about 5% of the total) and final spin (0.68) predict a ringdown spectrum whose first overtone matches the fundamental-mode inference to within the Kerr-family bound of 30%, and the area law — that the horizon area cannot decrease — is confirmed at 3.4σ across every analysis window and above 5σ in the most sensitive one; both are area-law-family statements this framework's exterior geometry inherits unchanged from general relativity, since nothing about the exterior solution differs. The damping-time precision of that same ringdown measurement (9%) is looser than the frequency precision (2%) by a factor of about 4.5, which matters because the damping time is exactly the observable a partially reflecting boundary would shift, while the frequency would not — so the ringdown measurement is currently most sensitive in precisely the channel where a reflecting-boundary effect would first appear, and has not yet reached the precision needed to see one.

The magnitude of $\mathcal{R}$ itself requires an equation of state for whatever replaces the interior — a computation explicitly outside this framework's scope (Section XII A) — but the qualitative structure, comb-like echoes and an exponential rather than linear bound, is a genuine, falsifiable prediction independent of that missing input.

VII. THE ANISOTROPIC SECTOR: LABELS AND CHARGES

Where the scale pattern is anisotropic, an order parameter, charges, and multiplets appear together, from the same source as the gauge connection of Section IV. A particle is a point defect in the scale vector field — an unresolvable knot — completely characterized by a triple: a threshold (its position on the scale axis, read as mass), a $\mathbb{Z}/2$ confined charge, and a $\mathbb{Z}$ winding charge, and nothing else. This is not an assumption about how many labels a particle may carry; the pattern having collapsed to a single vector (Section IV) leaves two topological invariants available in three spatial dimensions — a winding number (valued in $\mathbb{Z}$, from the vector's degree as a map to a sphere) and a monodromy class of the order parameter (valued in $\mathbb{Z}/2$, because the order parameter lives in a projective, not a linear, space). That these are the invariants a projective order parameter in three dimensions admits is a standard fact of algebraic topology, cited rather than proved in-repo; what *is* proved directly is the sharper, load-bearing half — that the confinement group is specifically $\mathbb{Z}/2$ and not some other value, which is what rules out the three-valued color identification below.

The $\mathbb{Z}/2$ nature of the confined charge is structural rather than a numerical near-miss: an earlier attempt to read this charge as color required three distinguishable values, and failed against the data by giving two. The order parameter's home is a projective space, and a projective order parameter simply does not have a three-valued invariant available to it — no adjustment of the

construction changes this.

Mass, in this picture, is not a third label alongside the two charges; it is the position on the scale axis at which unresolved content first becomes a distinguishable defect, so “mass” and “threshold” are the same statement read two ways rather than an independent property a particle additionally carries. This is also the precise sense in which there is no final, complete list of fundamental particles (Section II): resolving further along the scale axis always exposes more threshold structure above whatever has already been resolved, and the claim is not merely that undiscovered particles might exist but that, given an unbounded supply of ever-finer thresholds to resolve — itself a physical premise about the scale axis rather than a theorem derived from A1–A7 alone — the content strictly above any given resolution is never empty, so “the complete particle table” is not a question this framework’s content structure can answer even in principle, for any resolution one names.

A. The two charges are independent, and exactly one is confined

The winding charge and the monodromy (block) charge are logically independent: neither is determined by the other, since they are computed from different topological data — a degree and a monodromy class — of the same underlying configuration, and it is possible to exhibit configurations agreeing on one label while differing on the other. This independence is what makes “particle” a genuine pair of quantum numbers rather than one label in disguise, and it is what makes the statement “exactly one of the two is confined” non-trivial rather than automatic: confinement is a further, structural fact about the monodromy charge specifically — a configuration whose block labelling fails to close up after a circuit is not merely rare or energetically disfavored, it is not single-valued and so does not describe a state that can exist in isolation — while the winding charge carries no such obstruction and every winding number is freely realizable on its own. A defect paired with its own opposite always closes up and carries zero net charge in *both* labels simultaneously, independent of which block permutation either one carries, which is the framework’s account of why bound pairs (mesons, in the language of the withdrawn color identification) are always available regardless of the confined partner’s internal labelling.

The same rank-one collapse that limits particles to two labels also bounds the shape of the mass spectrum. If particle masses were drawn from a mixture of two or more structurally distinct populations rather than a single rank-one family, the squared coefficient of variation of the mass distribution satisfies $CV^2 \geq 1$, with equality exactly when the population is the single-family (rank-one) case this framework predicts; a mixture is not merely disfavored numerically, it is excluded whenever the measured CV^2 is below 1. The measured value, 0.875 (Ta-

ble II), sits below the bound and is consistent with rank one, though $0.875 \neq 1$ exactly, so the comparison establishes exclusion of mixtures rather than confirmation of the exact rank-one value — a distinction Section XIV records explicitly rather than eliding.

B. Gauge symmetry is the stabilizer of the pattern

Internal symmetry in this framework has no independent space to act on; the only structure at a point is the scale pattern across directions, so any symmetry that is not a motion of the substrate itself must be a relabeling of directions that leaves every local scale value unchanged. The group of such relabelings — the *stabilizer* of the pattern under the permutation action on directions — is not chosen; it is read off the pattern directly, and two directions belong to the same multiplet exactly when they carry the same local scale, so multiplets *are* the degeneracy classes of the pattern and multiplet sizes are the sizes of those classes. Where the pattern is injective (no two directions share a value) the stabilizer is trivial and there is no internal symmetry at all; where the pattern is fully isotropic the stabilizer is the entire permutation group.

This has a direct consequence for what a connection is. A transport between two scale patterns is a relabeling that matches them; where the target pattern is injective, that relabeling is unique — the connection is a function of the scale field alone, with nothing left to choose — and wherever it is not unique, the freedom in choosing it is exactly a coset of the stabilizer, meaning gauge freedom is not a separate structure layered onto the geometry but is *identical to* the degeneracy of the underlying pattern: gauge freedom exists at a point if and only if that point’s scale pattern is degenerate there, and its size is exactly the size of the degeneracy. A theory with no coincident scale values anywhere has no gauge freedom anywhere, not as a choice of gauge but as a consequence of the pattern being injective; internal symmetry is consequently never posited as an independent input; it is measured by counting coincidences in a single scalar-valued function per direction.

C. Particle lifetimes are scale ratios

Because energy scale and length scale are reciprocal (A3), a particle’s proper lifetime τ_0 and its lab-frame lifetime τ are related by the same ratio that appears everywhere else in this theory:

$$\tau = \gamma \tau_0, \quad \gamma = \frac{\varepsilon}{\varepsilon_0}, \quad (6)$$

the ratio of the particle’s energy scale in the lab frame to its energy scale at rest. Time dilation is accordingly not a separate relativistic postulate but the same scale-ratio statement that produces every other result in this paper,

evaluated for a moving particle; applied to the CERN muon storage ring measurement [14] this reproduces the measured lifetime to the precision of Table II.

D. Is there a second switch?

The framework predicts exactly one kind of label-opening event: an anisotropic pattern appearing where an isotropic one was, always in the direction of *refinement* — a multiplet only ever splits into smaller pieces as the scale is probed, never the reverse. The framework does not itself prove this refinement direction — only that a change of label structure requires a change of scale pattern — so testing the refinement direction against data is not circular.

We count exactly-conserved label *types* (not states) across the directly probed particle spectrum [15], spanning

$$\frac{E_{\max}}{E_{\min}} = \frac{1.3 \times 10^4 \text{ GeV}}{5 \times 10^{-11} \text{ GeV}} > 10^{14}, \quad (7)$$

fourteen orders of magnitude in energy. Over that range the electroweak multiplet only ever shrinks ($2 \rightarrow 1$, a strict refinement) and the color multiplet is unchanged ($3 \rightarrow 3$); no observed symmetry breaking has ever enlarged a multiplet, and no additional exactly-conserved label beyond the four already in the framework's inventory has been found. This is a genuine, if limited, empirical check: it can only detect *exact* degeneracies opening or closing, and fourteen decades, while large, is finite.

VIII. SCALE FLOW: RENORMALIZATION GROUP AND THE STRONG SCALE

A fiducial shift is a symmetry of the geometry (A5) but not automatically of the description of matter: to keep predictions fixed while the reference unit moves, the couplings used to describe matter must move to compensate, and that compensating motion is the renormalization group. Three structural facts follow from A5 alone, before any coupling is specified. First, the scale flow is forced to be a one-parameter group — shifting the fiducial by t and then by u is shifting it once by $t+u$, with no other composition law available. Second, self-similarity forces exponential solutions: any observable obeying the Callan-Symanzik-type equation $dO/dt = \gamma O$ satisfies $O(t) = O(0)e^{\gamma t}$ exactly, the same mathematical fact that produces exact de Sitter expansion in Section X — the same equation read at two different scales, not the same numerical constant in both places. Third, the effective-field-theory tower is forced: each excitation carries a log-threshold, the set of excitations available at a given log-energy is monotone increasing in that energy, and it is exactly constant between thresholds — the description does not merely happen to be piecewise constant, it is provably unchanged wherever no threshold is crossed. These

three facts fix the *shape* of any beta function forced by A5: piecewise constant, jumping only where the active content changes. They do not, by themselves, fix the coefficient at any single jump — that remains external input at this level of the argument.

The strong-interaction sector closes that remaining gap for one specific coupling. Reading a coupling not as a diagrammatic sum but as an accumulated *count* — how many resolvable defects a probe at a given log-scale has responded to — A5 forces that count's density to be constant in log-scale, since a scale-covariant world has no preferred log-scale for defects to bunch around. A constant density ρ integrated over an interval is linear, giving

$$g^{-2}(t) = g^{-2}(t_0) + \kappa\rho(t - t_0), \quad (8)$$

exactly the one-loop running form, with the one-loop coefficient identified as $b = \kappa\rho/2$ — a density of defects per unit log-scale in place of an imported group-theory coefficient (with $\kappa > 0$ retained as a hypothesis, not itself derived). Given that, asymptotic freedom is *equivalent* to $\rho > 0$: $g(t)^2 \rightarrow 0$ as $t \rightarrow \infty$ exactly when there are defects to resolve, so which sign of density gives a free theory is read off the count rather than fitted separately. The combination $\Lambda := \exp(t - g(t)^{-2}/(2b))$ is constant along this flow — a *scale-invariant* axiom system has produced a definite scale not by breaking the invariance in the axioms but by the solution of a scale-covariant equation secreting one, and A5 guarantees no choice of fiducial can shift it. Every hadron mass, read as $c \cdot \Lambda$ for a pure number c , then has a mass *ratio* c_1/c_2 independent of Λ and hence of every free input to the theory, in contrast to the overall scale Λ itself, which is measured rather than predicted, consistent with the one-free-unit statement running through this paper.

One caution is necessary about ρ , and it is the framework's own internal correction rather than an external criticism. The density entering the running above is *sector-restricted and signed*: a coupling responds only to the structures actually charged under it, not to every threshold in the spectrum, so it is emphatically not the same quantity as the raw threshold density used elsewhere (Section XIV) to bound the mass spectrum. A proposed cross-check — measuring this running-coupling density independently from the raw spectral count and comparing — was attempted and found not to be a test at all: the two counts differ by roughly a factor of two (0.864 counting all twelve Standard Model mass scales against 0.444 counting the six quarks alone), and within a single sector the relating equation has one free coefficient and one measurement, so it fixes the coefficient rather than checking anything. This is registered (Section XIV) as a proposal that was made and retracted, not as a claim that survives.

Measured against data, the simplest possible ratio law for the confined sector — an equal-ratio geometric tower of defect masses — was checked against lattice glueball ratios [16] (1.39–1.40 and 1.07–1.08 for the two lightest ratios, differing by more than 0.3) and **failed**, refuting

that specific hypothesis about the mass law rather than the parameter-free ratio construction itself (Section XIV records the retraction and why the underlying flow result survives it).

A. A second, sharper test: the shape of the threshold spectrum itself

The glueball comparison tests one specific alternative hypothesis using three numbers. A5's requirement that no log-scale be preferred makes a second, independent, and considerably sharper prediction using the entire spectrum at once. If thresholds are placed by a scale-invariant process — one of constant rate ρ per unit log-scale, with no preferred location for that rate to be higher or lower — the resulting point process has exponentially distributed gaps, and an exponential distribution has squared coefficient of variation exactly $CV^2 = 1$, with no free parameter. The geometric-tower hypothesis predicts the opposite extreme: equal gaps, hence zero variance, hence $CV^2 = 0$ exactly. The two predictions are as far apart as they can be, and this is a statement about the whole gap distribution, not about two or three selected ratios.

Taking the eleven consecutive gaps in $\ln m$ across the twelve distinct Standard Model mass scales, from the electron to the top quark, gives $CV^2 = 0.875$ ($CV = 0.935$) — inside the central 90% sampling range for eleven gaps under the scale-invariant prediction (roughly 0.33 to 1.64 in CV^2), and at the 63rd percentile of it, while the geometric tower's $CV^2 = 0$ lies far outside that range on the other side entirely. What this establishes, honestly stated, is the sharper half of the comparison: exclusion of the geometric tower is decisive, since 0.875 is nowhere near 0, while agreement with the scale-invariant value 1 is merely encouraging, since eleven gaps is a small sample and the twelve mass scales entering it were selected by hand (massless states excluded, neutrino masses unresolved). The same gap statistics fix the threshold density, $\rho \approx 0.86$ per e-fold — the same ρ_{all} that appears in the retracted cross-sector comparison above, now correctly understood as a property of the full spectrum's gap statistics rather than a quantity meant to match a sector-restricted running coefficient.

B. Label proliferation measures range, not structure

A description of content is exact between thresholds, as noted above, and gains exactly the thresholds it crosses when its range is widened, and no others: extending the window of validity from t to t' adds precisely the structures whose threshold lies in $(t, t']$, and if no threshold lies in that interval the description is unchanged. Two consequences follow directly, with no further assumption. First, the excess content a formalism must carry is monotone in the range it covers — widening the window at ei-

ther end can only add labels, never remove them, so the label count is a genuine function of how much log-range is covered. Second, no finite label set suffices at every scale: above any given scale there is always a threshold not yet crossed, so any fixed labeling scheme is eventually insufficient, not as a practical limitation but as a theorem.

This gives a direct account of why the Standard Model's label set is as large as it is, without touching whether its content is fundamental. A formalism required to remain valid across thirty orders of magnitude in energy must carry a label for everything that appears *anywhere* in that range, whether or not it is realized at the particular scale one happens to probe — most of those labels are, at any single scale, unrealized. This is exactly the shape of off-shell content, gauge redundancy, and species that exist only virtually: not evidence that the underlying structure is as rich as the label count suggests, but the unavoidable bookkeeping cost of a single formalism spanning a wide range. The claim is not that the Standard Model is wrong to carry the labels it carries, only that the size of that label set is telling an observer about the range being covered, and reproducing that particular count is consequently not a target this framework treats as meaningful.

IX. THE QUANTUM SECTOR

Promoting the commutative substrate to a non-commutative one and asking what changes gives the quantum sector. The commutators of the non-commutative scale algebra reproduce the Leibniz rule, the Jacobi identity, the power rule for derivatives, and the canonical commutation relation as theorems rather than postulates. Consequently, A3's scale-energy duality $\varepsilon \cdot s = 1$ continues to hold exactly only up to a correction of order $\hbar$ in the non-commutative sector — the classical relation is the $\hbar \rightarrow 0$ limit of a relation that is otherwise corrected, not an independent low-energy postulate.

Uncertainty is a general fact about symmetric (self-adjoint) operators on an inner-product space and does not require the scale algebra specifically: for any two symmetric operators A, B and any unit vector ψ ,

$$|\langle \psi, [A, B] \psi \rangle| \leq 2 \|A\psi\| \|B\psi\|, \quad (9)$$

by the Hermiticity of A, B , the triangle inequality, and Cauchy–Schwarz applied to the resulting inner products. Specializing to a commutator that acts as the scalar $i\hbar$ — which is what the canonical Poisson-bracket construction below produces for position and its conjugate momentum, once quantized — gives the Heisenberg relation $\Delta A \cdot \Delta B \geq \hbar/2$ as a corollary of the general inequality, not as an independently assumed postulate: the constant $\hbar$ enters only through evaluating the commutator, and the inequality itself is pure linear algebra.

Before the exact (Theorem 4) construction, a first-order deformation-quantization [17] version fixes what

quantization must mean here. On $A[\hbar]/(\hbar^2)$, a square-zero extension of the substrate ring, a star product is built directly from a biderivation P (a Leibniz bilinear pairing of two derivations, not itself required to be antisymmetric): $(f_0, f_1) \star (g_0, g_1) := (f_0 g_0, f_0 g_1 + f_1 g_0 + P(f_0, g_0))$. This product is associative and reduces exactly — with nothing approximated — to the ordinary commutative product at $\hbar^0$, and its commutator reproduces the *antisymmetric part* of P at $\hbar^1$: $\hbar^{-1}[f, g]_\star = \{f_0, g_0\}_P := P(f_0, g_0) - P(g_0, f_0)$, the Poisson bracket, so two quantities commute in the deformed algebra exactly when this antisymmetric part vanishes on them, and the commutative (gravitational) sector is recovered exactly as the locus where that bracket is zero — P itself need not vanish there, only its antisymmetrization. This is a first-order formal deformation, useful for identifying what non-commutativity must mean before asking whether it can be realized exactly; Theorem 4 below removes the formal parameter and replaces it with an honest additive-and-multiplicative shift map.

Theorem 4 ($\hbar$ as an exact scale step). *There exists a scale shift T — a map on the scale algebra additive and multiplicative, though not required to fix the multiplicative identity — such that T fails to commute with position by exactly one unit: $\Delta X := T(X) - X = 1$, exactly, not as a leading term in an expansion.*

This structure is never itself constructed from A1–A7 alone by the argument above; it asks only for a map with this weaker-than-ring-endomorphism property. We therefore exhibit an explicit model: the polynomial ring $\mathbb{R}[X]$ with d/dX satisfies A1 for $n = 1$, and the translation $X \mapsto X + 1$ is a ring homomorphism realizing T exactly, so the shift is a legitimate, exact, finite — not infinitesimal — translation along the scale axis. This shows the theorem is non-vacuous and the structure compatible with the axioms; it does **not** show that every model of the axioms must carry such a shift, so the identification of $\hbar$ with this step remains an identification, not a derivation, and is registered as such (Section XIV).

Theorem 5 (The quantum correction to curvature is purely antisymmetric). *The entire non-commutative correction to the deformation tensor is given by the commutator $[\sigma_i, \sigma_j]$ of scale generators, which is purely antisymmetric.*

This is the origin of Theorem 3: since the symmetric sector carries no quantum correction at all, the gravitational-wave radiative sector is untouched by non-commutativity to any order, while the antisymmetric (rotational/gauge) sector carries the entire quantum correction. This gives a genuine, structural prediction for *where* a quantum-gravitational effect could appear at all, even though its magnitude cannot be: the correction $[\sigma_i, \sigma_j]$ and the rotational label of Section IV are the same kind of object, both antisymmetric in the same index pair, so **quantum-gravitational corrections appear first in spin-geometry coupling, not in orbital dynamics**

— orbits are computed from the symmetric part, which is exactly the part this correction cannot touch. Stated honestly rather than left implicit: the size of the commutator is set by the same scale step that gives $\hbar$ (Section IX), so relative to the classical terms in the solar system this correction is utterly negligible, and nothing here predicts an effect within reach of current instruments — what is claimed is the structure of the leading correction and precisely which channel it can and cannot appear in, not a measurable number.

A. What this sector does not give, and a precise reason why

Everything above is kinematics: a commutation relation, an uncertainty bound, a correction to curvature. No dynamical equation of motion — no Schrödinger or Heisenberg picture, nothing playing the role of $da/dt = \frac{i}{\hbar}[H, a]$ — has been recovered, and the omission is not an oversight; it is forced, and provably so, by A6.

Any bracket is alternating, so whatever generates a derivation D through it — $Da = \{H, \cdot\}$ for some H — is thereby forced to conserve itself under D , since $DH = \{H, H\} = 0$ identically. This is the algebraic content of “a Hamiltonian conserves its own energy,” owed to nothing beyond the bracket’s antisymmetry. A6 gives the energy unit ε a nonzero drift along its own distinguished direction — $\dot{\varepsilon} = -\lambda\varepsilon \neq 0$, dissipation exactly as Section X uses it — so ε fails that self-consistency condition and cannot be the generator of its own drift, by no more than the two facts just stated. Standard, closed-system quantum mechanics is not exempt from this argument; it sits on the other side of it. Set $\lambda = 0$ and the obstruction is exactly what disappears: $\dot{\varepsilon} = 0$ holds trivially, and given any direction along which ε has been normalized to unit slope, the same bracket construction used above recovers ε as a genuine generator of the drift, for every element, not merely for one conjugate pair. The obstruction and its resolution are one fact, read on the two sides of A6, not a gap on one side and a separate result on the other.

That recovery, stated this way, is still only a fact about the *classical* Poisson bracket built from A1’s own commuting derivations — exactly what ordinary Hamiltonian mechanics already asserts, and by itself no more quantum than that. What earns the name is the same construction carried into the non-commutative deformed algebra itself: by the bridge already proved in Section IX (the star product’s commutator has vanishing classical part and $\hbar$ -coefficient exactly the Poisson bracket, at first order and exactly, not as a limit), the recovery above is identical to

$$[\varepsilon, a]_\star = \hbar \cdot d_p(a),$$

the exact, first-order form of the Heisenberg equation of motion $\dot{a} = \hbar^{-1}[H, a]$ with $H = \varepsilon$ — not its classical shadow, and not an approximation to it.

The honest summary is a converse pair rather than an admission: this framework does not recover unitary time evolution *because* it is open, and it recovers exactly the standard picture the instant it is not — which is a sharper claim than “dynamics has not yet been attempted,” and one this framework did not have available before A6 was stated precisely enough to check against.

X. DARK MATTER, DARK ENERGY, AND LIGHT

A. Dark matter: two routes, making opposite predictions

This framework offers not one account of dark matter but two, and it does not choose between them — an honest state of affairs rather than an oversight, since the two are logically independent and observationally distinguishable.

The first is a detection-threshold mechanism, requiring no branch-point argument at all. A detector is itself a physical system sitting at some energy scale ε_D ; by A3 that scale is the reciprocal of its spatial resolution, so a detector built to resolve small distances just *is*, in this framework’s vocabulary, a high-energy object — there is no separate fact about its resolving power beyond the scale it sits at. What it can register is not “an interaction” but an interaction that moves its own state by at least its threshold θ . An excitation whose own energy scale ε sits far below ε_D deposits a fraction $\varepsilon/\varepsilon_D$ of the detector’s scale — strictly positive, so the interaction genuinely happens, but driven below threshold whenever ε is small enough, so the observation genuinely fails. Gravity, meanwhile, couples to the total energy $N\varepsilon$ of a population of such excitations, which is blind to how that total is partitioned among them. A large multiplicity of very-low-scale excitations is consequently gravitationally loud and non-gravitationally silent *simultaneously and for the same population*, for any fixed gravitational signal and any fixed detector: the theorem constructs, for any target mass M and any detector threshold, a scale and multiplicity reproducing M gravitationally while sitting strictly below that threshold. On this route dark matter has thresholds and hence a genuine mass spectrum, and a sufficiently well-matched probe would eventually see a signal.

The second is the isotropic-trajectory mechanism of Section V: a region whose scale pattern is isotropic in direction still has curvature, since magnitude inhomogeneity, not direction inhomogeneity, sources gravity, so such a region gravitates with no anisotropic order parameter and hence no electromagnetic or confined charge (Section VII) at any scale whatsoever — not a small coupling waiting to be resolved, but no label to couple to in the first place. On this route dark matter has no thresholds and hence no mass spectrum and no species to enumerate: it is one structural condition on the scale pattern,

not a family of candidate particles with different masses and couplings, and it forbids exactly what the first route permits — a charge-type self-interaction, since that requires the directional order parameter this route’s matter provably lacks.

The two routes make **opposite** predictions and the framework states this as a fork rather than resolving it: the low-scale route gives a coupling that is small but strictly nonzero at every finite separation, hence a spectrum in principle detectable by a sufficiently sensitive or well-matched probe; the isotropic route gives a rotational label that is identically zero at *every* scale, hence collisionless behavior and no spectrum to find by any amount of improved sensitivity. A null result at ever-improving direct-detection sensitivity favors the isotropic route; a spectral feature or a self-interaction signature favors the low-scale route; the framework hosts both constructions and does not adjudicate between them, which is a weaker but more honest position than committing to either.

B. The direction of time from finite resolution alone

The entropy ordering used in Theorem 1 rests on an H -theorem that uses nothing about scale in its own proof, which is precisely what makes the later identification with the scale-drift arrow a substantive fact rather than a definition chasing its own tail. The construction is elementary: let X be a finite set of microscopic configurations and $\pi : X \rightarrow Y$ a resolution map, generally non-injective because observation happens at finite resolution. The *saturation* of a subset $S \subseteq X$ under π is the set of all configurations π cannot distinguish from something in S ; a configuration set is always contained in its own saturation, so coarse-graining can only add configurations, never remove them, giving $|S| \leq |\text{sat}(\pi, S)|$. Microscopic evolution, being a bijection (the substrate is reversible — nothing in A1–A7 privileges a direction of time at the microscopic level), preserves the cardinality of any subset exactly. Composing one step of reversible evolution with one step of coarse-graining therefore cannot decrease the number of configurations compatible with what is observed: entropy, read as the log of that count, is non-decreasing by counting alone, with no statistical assumption beyond finite resolution — and once it has strictly increased at one step, no entropy-non-decreasing law can undo the evolution, which is the precise sense in which a reversible microscopic law produces an irreversible macroscopic one, matching the status of the Boltzmann equation.

This also gives the framework’s structural reading of Sakharov’s third condition for generating any asymmetry at all: departure from thermal equilibrium. In a closed system that departure is a fine-tuned episode that must be arranged; under A6 the universe never stops dissipating, so the condition is satisfied permanently rather than accidentally — strict entropy production is itself

a departure from equilibrium, since every observable is stationary in equilibrium by definition, so an open, dissipating universe satisfies Sakharov’s condition at every moment it produces entropy at all. This is deliberately modest: it establishes that the necessary condition holds throughout this framework’s history, not that any particular observed asymmetry (baryon or otherwise) follows from it.

C. Dark energy is the dissipation rate, squared

Cosmic expansion is, in this framework, not the motion of anything through space but the running down of the total dimensionless energy scale ε , read through the duality A3 as a growing length scale. Dissipation at a constant fractional rate λ has a unique solution, $\varepsilon(t) = \varepsilon_0 e^{-\lambda t}$ — no ansatz is assumed, the exponential is forced by the differential equation alone — and the corresponding scale factor $a = 1/\varepsilon$ is exact de Sitter expansion, $a(t) = a_0 e^{\lambda t}$, with constant Hubble rate $H = \lambda$ and $\ddot{a} = \lambda^2 a > 0$: the expansion accelerates for any nonzero dissipation rate, with no repulsive substance posited anywhere. “Dark energy” is accordingly not a substance with negative pressure; it is this dissipation rate, and matching the observed density gives $\Lambda \propto \lambda^2$ (specifically $\rho_\Lambda = 3\lambda^2/8\pi G$ once λ is identified with the asymptotic Hubble rate) — the *square* of the dissipation rate, since Λ enters through the acceleration $\ddot{a}$ rather than through λ itself, not the dissipation rate directly.

Pure exponential dissipation is the sharpest and most falsifiable case, not the only one available: generalizing $\dot{\varepsilon} = -\lambda\varepsilon$ to a power law $\dot{\varepsilon} = -\lambda\varepsilon^q$ for a single dimensionless exponent q fixes the equation of state exactly, $w = (2q - 5)/3$, invertibly, $q = (3w + 5)/2$. The case $q = 1$ is exactly $w = -1$ with no freedom at all — pure exponential dissipation — and it is the only value of q that gives $w = -1$, not one case among several that happen to agree. This is the framework’s sharpest exposure in this sector: if $w = -1$ is excluded by observation, pure exponential dissipation is dead, and the surviving theory must have $q \neq 1$, read off the data by the inversion formula. What the power-law generalization does *not* provide is a mechanism for w to *evolve*: for any fixed q , w is still a constant, since q is a single dimensionless exponent rather than a function of time, so DESI’s [18] preference for an evolving w_0, w_a remains a tension with the shape of the construction itself and not merely with the value $q = 1$ — a constant-but-shifted w is available to this framework in principle, a genuinely time-varying one is not, which is why the DESI tension of Section XII is correctly read as a tension with A5’s constancy requirement rather than with a single fittable number.

D. Light is scale-blind, but redshift is not

Reciprocity $s_t \cdot s_r = 1$ (Section VI) holds pointwise, not only in vacuum, so the null cone defined by $g_{tt}g_{rr} = -1$ is invariant under any global rescaling $s \mapsto u \cdot s$ permitted by A5. Since a photon’s trajectory is defined purely by the null condition, its propagation is insensitive to the overall scale — the light cone is scale-blind by the same fiducial covariance that makes any single scale unobservable. This gives photon dispersion identically zero at every energy, a stronger and structurally distinct statement than merely “dispersion smaller than currently measurable,” and it is the basis for the Fermi-LAT comparison of Table II [19].

This appears to contradict the cosmological picture directly above: if light cannot see the scale field at all, how can a redshift measurement ever register cosmic dissipation, which is exactly a change in that field? And Section VI’s local-covariance result sharpens the puzzle rather than resolving it, since it proves a spatially constant cosmic drift changes *nothing* measured locally. Both results are theorems, so both are true, and the appearance of contradiction is a sign that two different things were being called by the same name. They were. A null vector has no length, but assigning it an *energy* is not a property of the vector alone — it is the pairing of the vector with an observer’s own direction, and an observer is made of matter, which (unlike light) does carry the scale. The *causal structure* light defines is scale-blind, exactly as above; the *energy* an observer assigns to a light ray is not, because assigning it requires a scale-carrying observer in the first place. Redshift is consequently not a property of light at all: it is the *ratio of two observers’ scales*, $(u_1/u_2)^2$, with light contributing nothing to that ratio beyond carrying the comparison from one event to another. This simultaneously resolves the apparent tension with local covariance: a redshift measurement is not a local measurement of the drift, it compares the scale at emission with the scale at absorption, and that is a scale *difference*, which A5 declares physical, while local covariance concerns what a single observer measures at one point — cosmology observes exactly the quantity the axioms permit and no more.

XI. QUESTIONS ONLY THIS FRAMEWORK CAN ASK

Everything so far has answered questions posed elsewhere: does the theory give Schwarzschild, does it give a graviton-like sector, does it give an area law. That is the right first test, and every section above has run it. It is not the right last one, because a framework built on different primitives should be able to *ask* things the usual vocabulary cannot pose at all, and whether it can is where it either earns its keep or is exposed as a paraphrase of existing physics in new notation. Seven such questions follow from the primitives already in use — a

scale that is a unit, a pattern that collapses to one vector, a single global drift, the antisymmetric wedge, a threshold density, the isotropic/anisotropic dichotomy, and one commutator — each pushed for what it demands to know rather than for what it was built to answer.

Separation is not primitive; non-parallelism is. With no space as a primitive, “two places” cannot be assumed — it must be constructed, and the wedge is the only construction available (Section II). The question this opens, unavailable in a theory that starts from a manifold: what is the observable that measures *degree* of separation, if not a distance? The wedge answers it, and being anti-symmetric, the framework’s own separation is necessarily *oriented* — a signed area, not a length, which is not in the standard vocabulary at all.

There is one switch, not one per interaction. The isotropic locus has no rotational label, no charge, no multiplet, and no curvature simultaneously, because all four have one source (Section V). The framework’s own question: the usual account has separate energy scales at which separate interactions become distinct; here there is one boundary and then refinement. So — is there any label whose appearance is *not* a refinement of an already-anisotropic pattern? A genuinely independent switching-on would falsify the single-source claim, and it is a question no other framework needs to ask, because no other framework has only one switch to defend (Section VII D runs this test against fourteen decades of data).

Content belongs to intervals, never to a single scale. A window of zero width resolves nothing; with threshold density ρ , a window of width Δ contains $\rho\Delta$ structures, vanishing with the window. “What exists at energy E ” is consequently malformed as a question in this framework: what exists is a property of a scale *interval*, never of a point. The complementarity this produces — sharpness in scale opposed to richness in content — is not Heisenberg transplanted; the conjugate pair is scale-sharpness against content-count, and the smallest window in which a single structure is unambiguous is exactly $1/\rho$.

There is no global field equation, and A6 is why. A field equation can be written only if its source obeys a cyclic conservation law, and A6 says the universe’s global source is not conserved — so there is no field equation for the universe as a whole, only local ones, which survive because local covariance survives global dissipation (Section VI). The framework’s own question: if nothing global is being solved, what fixes the large-scale history at all? The only candidate is A5 itself, a symmetry rather than a dynamical law — the large-scale history is fixed by what is *unobservable*, not by what evolves, and whether that is sufficient is an open question this framework poses that general relativity, built around a global field equation, cannot.

The quantum-gravity experiment is an ordering test, not a particle search. The entire quantum correction to geometry is a commutator of scale gradients (Section IX), and a commutator is an order discrepancy: the native experiment is not “detect a graviton” but “compare the

scale along two directions, in both orders, and look for a discrepancy.” The magnitude of that discrepancy is set by the scale step and sits far below anything currently measurable, stated here rather than hidden; what is distinctive is that the observable this framework proposes is an *ordering*, not an amplitude, and no particle needs to exist for the question to be well posed.

Defects change parity in pairs, with no coupling constant. The confined charge’s $\mathbb{Z}/2$ label composes multiplicatively, so the total parity of any collection cannot change under binding or splitting. This is not a selection rule imposed on a Lagrangian — there is no Lagrangian here — it is arithmetic in $\mathbb{Z}/2$: is there any process that changes total parity? The framework says no, categorically, which is a cleaner statement than a selection rule derived from an assumed symmetry, since there is no symmetry being assumed and hence nothing to break.

There is no maximum density. A ratio of scale units is itself a unit — never zero, never infinite, but unbounded in magnitude — and matter, in the anisotropic sector, *is* anisotropy of the scale, so there is no maximum matter density, no Planck-scale density cutoff, and no scale at which the description must be replaced by a qualitatively different kind of theory. This is the same statement as the absence of a minimum length (Section VI), seen from the matter side rather than the geometry side, and it is the framework’s sharpest structural disagreement with programs that posit a fundamental cutoff.

None of these seven is a translation of a question asked elsewhere; each follows from a primitive the standard vocabulary lacks. Their honest status varies and is stated here rather than left implicit: the one-switch question and the parity question are testable now in principle; the interval-content complementarity is already tested through the resonance-width bound of Section VIII; the no-maximum-density question is testable only through its contrapositive (a confirmed minimum length would falsify it); the separation, global-history, and ordering-test questions are conceptual claims with no current measurement attached. None of the seven is a prediction of a number.

XII. CONFRONTATION WITH DATA

Table II is the single scoreboard of this paper, and its rows are not all making the same kind of claim, so the horizontal rules inside it separate five genuinely different epistemic categories rather than merely grouping topics. The first block is where the argument closes completely: light bending and perihelion advance follow from the axiom chain with no free parameter beyond the one dimensionless unit any such theory must carry, so agreement to the quoted precision is a real success and not a fit. The second block is arithmetic, not discovery — muon dilation and gravitational redshift are the scale-ratio reading of A3 evaluated on a known configuration, included be-

cause the framework should get elementary cases right, not because getting them right is informative on its own. The third converts a measured number into this framework's own units, using the one free constant λ that every version of the theory carries; agreement there tests the conversion, not the theory's structure. The fourth block is where the framework is actually exposed: live measurements that have not yet settled, where a wrong answer would cost something. The fifth is the sharpest block of all — four cases the framework excludes using its own machinery, listed not as failures avoided but as evidence the theory is capable of being wrong internally, which a construction with enough free parameters to fit anything could never produce.

Quantity	This framework	Measured	Status
<i>Derived from the axiom chain (plus $A6'$); certified to the precision quoted</i>			
Solar light deflection	1.7515''	$1.7509 \pm 0.0002''$	agrees
Mercury perihelion advance	42.989''	$42.98 \pm 0.04''$	agrees
PPN γ	1 (exact)	$ \gamma - 1 \lesssim 2 \times 10^{-5}$ [20]	agrees
<i>Arithmetic check of the A3 reading, not an independent prediction</i>			
Muon time dilation	64.43 μ s	$64.378 \pm 0.026 \mu$ s	agrees (10^{-3})
Pound–Rebka	2.455×10^{-15}	$(2.46 \pm 0.03) \times 10^{-15}$	agrees
<i>Converted; λ itself is measured — the one free unit</i>			
Dark energy density	$5.845 \times 10^{-27} \text{ kg/m}^3$	$5.86 \times 10^{-27} \text{ kg/m}^3$ [21]	agrees
<i>Currently live tests</i>			
w non-evolution (forced by $w = -1$, no running A5)		DESI: $w_0 > -1$, $w_a < 0$	tension , 2.8–4.2 σ
$f_0(500)$ pole Γ/m	< 1.157	CCL 1.234 [22] / GKPYP 1.090 [23]	open
Photon energy-dependent delay	exactly zero	$E_{\text{QG}} \gtrsim 7.6 E_{\text{Pl}}$	passes, distinct prediction
Gravitational wave speed	exactly c , no dispersion	$ \Delta v/v \lesssim 10^{-15}$	passes
GW polarization count	exactly 2, no breathing mode	network polarization tests	passes
Dipole radiation	identically 0	double pulsar $< 1.3 \times 10^{-4}$	passes
Ringdown echoes	reflectivity $\neq 0$	GW250114 residual = noise	open, under pressure
Ergoregion instability	reflectivity bound uncomputed	GW241011 $\chi_1 = 0.64$, long-lived	open, tighter pressure
Ultralight scalar	no such slot	$10^{-13} - 3 \times 10^{-12} \text{ eV}$ excluded	compatible
Area-law analogue	scale ratio never decreases	consistent with observation	passes (not discriminating)
Mass spectrum CV^2	$= 1$ (rank one; mixtures need ≥ 1)	0.875	partial
Exactly-conserved unconfined charge	exactly one	electric charge, $B-L(?)$	undecided, awaits $0\nu\beta\beta$
Black hole entropy	$\propto \ln R$	—	unmeasured (area law also unmeasured)
<i>Excluded by the framework itself — the most informative rows</i>			
Equal-ratio mass tower	$\text{CV} = 0$	0.935	excluded
Isotropic-sector-only deflection	0.8758''	1.7509''	excluded (factor 2)
Isotropic-sector-only precession	14.33''	42.98''	excluded (factor 3)
Color = block charge	$\mathbb{Z}/2$ (two values)	needs three values	excluded (structurally)

TABLE II. The complete numerical scoreboard.

Read against the five-fold structure above, the honest summary is that the first three blocks are not really under test — they close by construction or by simple substitution — and the entire empirical weight of this framework currently rests on the fourth block. One live tension and one live open question there are not fatal — both sectors are under active measurement — and the last four rows are options the framework itself rejects; these are more informative than the passing rows because they demonstrate the framework is capable of being wrong internally. On timescale: even a seesaw-based lepton-number-violating model predicts a $0\nu\beta\beta$ half-life far above current exclusion limits [24], so this is a genuine but slow-fused test.

A. Equation of state is out of scope

The magnitude of the ringdown reflectivity $\mathcal{R}$ (Section VI G) requires an equation of state for the interior structure replacing a horizon, which varies enormously between candidate compact objects and which this framework does not attempt to compute — this is explicitly outside its scope, not a gap it intends to close. What the framework *does* deliver without an equation of state is the qualitative echo/ergoregion structure of Section VI G and the comparison between the linear and exponential strength of the two observational bounds.

XIII. HOW TO FALSIFY THIS FRAMEWORK

The three tables below are ordered by how much damage a positive observation would do, not by topic, and the ordering is the point: moving down the tables trades the severity of the wound for the precision of the diagnosis. Table III lists four observations that would kill the entire framework outright, because each is forbidden by one of the axioms directly rather than by a computed number that could in principle be retuned — there is no adjustable constant anywhere in this construction that absorbs a detection of any of them. Table IV lists observations that would kill one sector while leaving the rest of the chain untouched, which is possible only because the sectors are logically separable consequences of the same axioms rather than independent add-ons patched together. Table V is different in kind from the first two: nothing in it would falsify anything by itself, but each row would force an explicit admission that a quantity this framework cannot currently compute has become the quantity an experiment is actually probing.

Observation	Kills	Status
Kills the whole framework		
w evolves with redshift (5σ , dataset-consistent)	A5, dimensionlessness, all geometry	DESI 2.8–4.2 σ , but BAO alone self-consistent
Energy-dependent photon delay, any order	A3 + light-cone scale-blindness	$E_{\text{QG}} \gtrsim 7.6E_{\text{P1}}$, winning
GW dipole radiation	the conservation theorem, hence Bianchi	double pulsar $< 1.3 \times 10^{-4}$
GW breathing mode	A5 (a breathing mode is a fiducial shift)	tensor favored over scalar/vector/mixed

TABLE III. No adjustable parameter absorbs any of these four: a detection is fatal, not a call for retuning.

Below this tier, a positive detection stops being a global catastrophe and starts being informative about exactly which piece of the chain failed, because the four axiom-level results above do not have to fail together.

Observation	Kills	Status
Kills one sector		
A minimum length exists	horizon, singularity max-density results	sin- same bound as photon and dispersion
$f_0(500)$ confirmed	pole Section VII's re- $\Gamma/m > 1.157$ solvability bound	CCL above, GKPY below
$B-L$ and conserved	gauged the “exactly one exactly charge” result	“exactly one $0\nu\beta\beta$ not seen, slow fuse
A label opens refinement	Section VII D's that is not a single switch	no counterexample in 14 decades
Total-parity-violating process	the $\mathbb{Z}/2$ arithmetic	none known

TABLE IV. Each row is a local consequence of one construction rather than the axiom chain as a whole: a positive result here diagnoses which sector broke without impugning the others.

The last tier is not falsification at all in the ordinary sense — every row in it is compatible with everything proved so far — but each names a place where the framework has already admitted, in Section XII A, that it cannot compute the number an improving measurement is closing in on.

Observation	Exposes	Status
Forces an admission of what cannot be computed		
Rapid spin- horizon- ning compact absence object decays quires small via ergoregion reflectivity, instability uncomputed		GW241011 $\chi_1 = 0.64$, re- long-lived; tightest cur- rent pressure
Ringdown echoes seen	same, re- versed: predicted to exist, am- plitude not predicted	null searches so far
BH entropy $\propto$ the area, not $\ln R$	log- entropy result	entropy itself unmeasured

TABLE V. None of these rows falsifies the framework by itself; each marks where an improving measurement is closing in on a quantity Section XII A already registers as uncomputed.

A7 cannot be falsified by this table. It is a commitment about what counts as an observation; anyone may reject it and obtain a different k , and no experiment adjudicates that choice — which is precisely why it was promoted to an axiom rather than left an informal judgment. The one free unit is likewise not falsifiable: it is the scale in which predictions are expressed, not itself a prediction.

What does *not* constitute falsification: the absence of a Standard Model particle table (Section VII argues this was never a goal); the absence of computed hadron masses (the same free unit again); and, emphatically, agreement with general relativity in the gravitational-wave sector — Theorem 3 proves the symmetric sector agrees at every order, so no discrepancy is expected there in principle.

XIV. METHODS AND PROVENANCE

The argument was developed and checked entirely inside the Lean 4 proof assistant: every theorem stated informally above has a corresponding Lean statement and machine-checked proof, and `#print axioms` run against the complete development (650 theorems) reduces every one of them to `propext`, `Classical.choice`, and `Quot.sound` — the standard logical axioms of Lean’s own mathematical library, and nothing else; a companion script cross-checks that this audit, the paper’s own citations, and the theorem counts quoted in it have not drifted apart. That guarantee is about validity, not truth: it says an argument follows from stated premises, not that the premises hold of nature, and it says nothing about whether a claim made in English next to a theorem is a faithful reading of what the theorem actually says. The check against the second failure mode was adversarial external audit, repeated across several rounds, each

of which found genuine errors — some in the mathematics, most in the prose wrapped around it. What follows is the resulting record, kept rather than smoothed over, because a theory’s history of correction is evidence about how it was checked. The complete formal development and its supporting tooling are available in the project repository, along with a companion technical report.

Retracted (4). A geometric mass-tower law, later shown to be an unproved assumption rather than a derived consequence, and refuted by lattice glueball data under that assumption; a color identification that required three values where the projective order parameter structurally provides only two; a proposed cross-sector consistency test comparing two quantities that were not, on inspection, the same quantity; a codimension analysis that described a circle-valued rather than directional scale.

Corrected in scope (6). In each case a theorem survives and an earlier reading attached to it does not. Commuting derivations were read as a hidden flatness assumption; killing every derivation leaves curvature untouched, because flatness lives in whether the *connection*’s transports commute, not whether the underlying derivations do (Section IV). Directions were read as the algebra itself, with n its dimension; $\dim \mathfrak{so}(3, 1) = 6 \neq 4$, and directions are the *defining representation* the algebra acts on, not its own coordinates (Section IV). The interior of a multiplet was read as unconditionally flat; the result needs a per-direction scaling assignment as a hypothesis, since real rank one by itself supplies only one vector and one scaling overall (Section V). A uniaxial pattern was read as carrying no rotation, creating an apparent dilemma with the particle sector’s own labels; the theorem is about a *homogeneous* pattern, not the number of axes, and the dilemma dissolves once the misreading is removed (Section IV). The scaling and rotation channels were read as independent; that holds only for the scalar version of the theory, and the directional scale makes the rotation sector *generated by* the scaling sector instead (Theorem 2). The $f_0(500)$ pole was read as violating the resolvability bound; the previously cited value was not a settled determination, and the genuine range straddles the bound between two independent extractions — open, not violated (Section XII). Two further identifications are registered alongside the six rather than counted separately, since each softens a claim already listed rather than retracting a new one: real rank one’s step from “one drift functional” to “one independent scaling direction” is a physical identification, not a formal deduction (Section IV), and the measured mass-spectrum $CV^2 = 0.875$ excludes mixtures without confirming the exact rank-one value (Section VII).

Five of the six share one root cause: scalar reasoning applied where the structure is directional. The diagnostic that catches it: whenever a claim mentions a count — one axis, one scale, one direction — ask a count of *what*; value and direction are never the same count.

Cited, not proved (2). The classification of real-

rank-one real simple Lie algebras invoked in Section IV, and the relevant homotopy facts about projective spaces used in the particle-charge construction of Section VII. If either citation fails, the weaker statement proved before invoking it still stands; only the identification built on top does not.

Assumed, registered rather than smuggled (4). $A6'$, the scale response law, external to the seven axioms; a uniform-density assumption needed for one dimension-counting step in Section IV; $A7$ itself, promoted to an axiom rather than left implicit; and $A6$'s generalization from a statement about the total energy scale alone to one about the substrate's full multi-directional drift, used throughout but not itself proved from the narrower axiom as formalized.

Structures now witnessed (6 of 6). Six structures carrying load-bearing theorems — most importantly the scale-shift structure carrying Theorem 4 — were, as found by external audit, never constructed from $A1$ – $A7$ anywhere in the original development, making the associated theorems true but conditional. All six are now witnessed by explicit models: the polynomial-ring construction of Section IX witnesses the scale-shift structure itself and a cosmic-history example; the remaining four are witnessed by separate, unrelated concrete models — a matrix trace, an adjoint action on a general ring, a real-valued translation flow, and a constant function — chosen for whichever structure each was needed for, not by one uniform construction. This shows the structures are *compatible* with the axioms, not that the axioms *force* them — the gap between compatible and forced is real and is not closed by exhibiting a witness.

What rests on the axioms alone. The scale/rotation split, the codimension-one signature split, the bookkeeping form, the coupling (Theorem 2), real rank one (subject to the physical identification above), conservation from Bianchi, the full gravity chain through $1.7515''$ and $42.989''$ /century, the label structure of the anisotropic sector, every statement in Section XI including the one-arrow-of-time identity (Theorem 1). $\hbar$ as an exact scale step is deliberately absent from this list: it is witnessed, not forced, exactly as stated above.

Use of AI tools

AI language-model tools assisted in writing the Lean 4 formalization and in editing this manuscript. Every theorem so produced was checked by the Lean 4 kernel independently of how it was written, and all resulting code and text were reviewed by the author before inclusion.

XV. CONCLUSION

Taking scale rather than spacetime as primitive, a single chain runs from axioms to numbers. Scale must be directional, since a scalar version forbids the Schwarzschild

metric outright. Once directional, the scale–rotation coupling is forced by the meaning of the two words: scaling is symmetric, rotation is antisymmetric, and the bracket of two symmetric operators is necessarily antisymmetric. The coupling, together with the drift being a single functional, gives real rank one, hence the Lorentz family, hence a fixed dimension with three spatial directions set by $A7$ — a commitment about what counts as an observation, not a free parameter. The metric connection and the gauge connection turn out to be the same commutator, read under two different questions. The isotropic/anisotropic branch determines whether labels exist at all, itself a scale-dependent fact, giving this framework's account of why interactions separate as energy falls. Curvature is scale deformation, the renormalization group is the same drift read as a running coupling, $\hbar$ is a step along the scale axis, and the entire quantum correction to geometry is purely antisymmetric. Dark matter is the isotropic trajectory of the same field that produces ordinary gravity, and dark energy is that field's own dissipation rate. The gravity sector closes quantitatively at $1.7515''$ and $42.989''$ /century with nothing assumed in between.

These are not metaphors; every claim stated as a theorem in this paper has an independent, complete proof, verified end to end by machine — 650 checked theorems, no admitted gaps, and no hidden axioms beyond those of standard mathematics. But the statement that this framework reproduces known physics cannot currently be made without qualification: the scoreboard carries one live tension, DESI's 2.8 – 4.2σ preference against w non-evolution, which is a tension with $A5$ itself and not with a fittable parameter, since no version of this framework admits an evolving w ; and one live open question, the side of the resolvability bound on which the true $f_0(500)$ pole lies. Nor does the framework explain the Standard Model — Section VII argues that was never the goal — and Section XIV records four retracted claims, six claims corrected in scope, two citations of results not reproved here, and four assumptions registered rather than hidden, along with one recurring scalar-for-directional error caught five times.

What a fully verified argument can establish is one thing above all: that these conclusions follow from these axioms, with no gap in between. Whether nature adopts the axioms is a separate question that this paper does not, and should not be read to, answer — which is exactly why the scoreboard and the register are given in full rather than summarized away.

A. What this paper does and does not claim

Two questions were left open earlier for exactly this point. Section II asked what status to give a framework built this way — whether the axioms should be read as a claim about the world or as the tightest set of commitments found so far that reproduce it — and asked, more

sharply, whether scale is what actually exists or only the leanest set of assumptions available. This paper does not choose between them, and the choice is not a detail deferred to a sequel: it is the oldest disagreement in the philosophy of physical theory, between a realist reading on which a successful theory’s posits are what the world is made of, and a relationalist or instrumentalist reading on which a theory’s success licenses belief in its structure without licensing belief that its posits are furniture of reality [25]. Nothing argued here adjudicates it, and nothing the machine verification does could: Lean checks that theorems follow from axioms, not that the axioms are true of anything. What can be said is only that the axioms were chosen for how little they assume, not for how well they match a prior intuition about what exists, and a reader is free to take the derivations that follow as evidence for scale’s reality, as evidence only for an economical bookkeeping scheme, or as neither, without altering a single proof in this paper.

The same restraint applies to this paper’s relationship with programs it does not reach. As Section I states and this section now confirms without qualification: this is not offered as a competitor to loop quantum gravity, string theory, or causal set theory, and it does not reach the questions — a matter content, scattering amplitudes, a working route to quantized gravity in either the loop or the string sense — that make those programs what they are. What it offers instead is a single, narrow, machine-checked existence proof that geometry, gauge structure, and a testable gravitational sector can be derived together from a smaller set of commitments than is usually assumed necessary. An existence proof of that kind is compatible with every other program eventually turning out to be the more fruitful one.

B. Open questions this framework raises but does not settle

Spacetime has usually been treated as a thing. General relativity curves it, loop quantum gravity and its relatives quantize it, causal set theory discretizes it — each treats spacetime as a substance requiring a verb. This paper does not argue that habit of thought is wrong; it shows only that a smaller move — reading spacetime off the pairwise differences of one scalar-valued field, with no independent metrical input (Section II) — reproduces enough of what a substantival spacetime is asked to do that the substance becomes optional rather than required. Whether physics is better served treating spacetime, or scale, or whatever eventually plays their role, as something to be discovered or as a derived bookkeeping device is the substantivalist–relationalist debate under a new name [25]; this paper adds one data point without resolving it.

A theory is always a description of a slice, not of everything at once. Section VIII B proved this for one construction — a description valid over a range of scales

necessarily carries labels for content not resolvable at any single scale within that range, and no finite label set is ever adequate at every scale — but the result reads more naturally as an instance of something broader: that effective field theory may not be a staging post on the way to a final, all-scale theory but the only kind of description a scale-structured world can support at all. If that is right, a complete, finite inventory of nature’s content is not merely hard to obtain; asking for one is asking a single slice for what only an unbounded sequence of slices could give. Whether that generalization holds beyond the one construction it was proved for is left open here.

Precision is always precision at a resolution. A5’s fiducial covariance and the scale-sharpness/content-count complementarity of Section XI together say that asking what a quantity equals to infinite precision, independent of the scale at which the question is posed, is not a question this framework declines to answer — it is a question the framework has no vocabulary to pose, since infinite precision would be a point on the scale axis, and a point carries no content. Physical precision, on this reading, is approached and never reached, not because instruments are imperfect but because “reached” is not a coherent target for a quantity defined only relative to a resolution. Whether that is a fact about this construction specifically or about physical measurement generally is, again, an open question.

A theory earns the name only if it can lose. Popper’s demand that a scientific theory be falsifiable rather than merely confirmable [26] is not treated here as a slogan: Section XIII names four observations that would kill this framework outright with no adjustable parameter available to absorb them, and Section XII’s scoreboard leaves the DESI tension and the $f_0(500)$ question standing rather than retuning anything to make them disappear, because nothing in this construction is left free to retune. A framework flexible enough to accommodate any observation has not survived contact with data; it has not yet made contact with it. Whether this framework’s rigidity is a virtue, or merely a consequence of how little it currently claims — a theory with no adjustable parameters is easy to falsify in part because it is not yet trying to explain very much — is left for the reader to weigh against Section XII’s scorecard rather than settled by the author here.

What, in the end, is theoretical physics for? Elsewhere I have argued [27] that theoretical physics should be held to three mandate axioms: that its mandate is to describe, explain, and predict nature; that its subject matter is accessible, or at least could be approximated with arbitrary precision, with an objective that is eternal and invariant; and that it should exclude self-referential propositions or statements about its own constitution, on the grounds that such propositions do not increase productivity, creativity, or other benefit to humanity. Read against those axioms, this framework’s own limits are directly relevant rather than incidental: the impossibility of a finite universal label set (Section VIII B) and the never-reached

character of precision noted above are close to the shape the second mandate axiom anticipates — an eternal, invariant objective approached asymptotically rather than attained in finite time is plausibly what “approximated with arbitrary precision” has to mean if it is to mean anything achievable at all. Whether an eternal, invariant objective exists to be approximated in the first place — whether a final description of nature, or a genuine unification, stands at the limit of that asymptotic approach, or whether no such limit exists — is not a question this paper can answer, and it is not obviously a question physics as a discipline can answer either, since the answer would itself have to be established by the very activity in question. Two literatures bear on this without settling it. Gödel’s incompleteness theorems [28] establish that any sufficiently expressive formal system contains true statements it cannot prove from within itself; whether that result has any bearing on physical theory, as opposed to the formal axiomatic systems it was proved for, is itself contested — Hawking read it as grounds for doubting that physics could ever be complete in the sense of a finite, closed set of laws deciding every question [29], Dyson has speculated in a similar direction [30], and Barrow’s more careful treatment concludes the analogy is suggestive without amounting to a theorem about physics itself [31]. This paper takes no side in that dispute; it only notes that this framework’s own structural results — no finite label set, no maximum density, content that is never complete above any given scale (Sections VII, XI) — have exactly the flavor the incompleteness literature describes, without claiming the resemblance is more than that. And if no final description exists, the third mandate axiom’s exclusion of self-reference suggests that asking what physics is *for* is, in that case, a question for philosophy of science and for the scientific community collectively rather than one this paper’s conclusion is positioned to settle. Whether this framework meets the

mandate at all — whether deriving a consistent, checkable structure from seven axioms, without yet explaining the Standard Model, counts as describing, explaining, and predicting nature, or only as a preparatory exercise for a theory that might someday do so — is left, deliberately, open.

What counts as an observation is itself a choice, not a discovery. A7 is stated in Section III as exactly this: a commitment about what a single observation is, not a claim about the world, and Section XIII records that no experiment can adjudicate it. That this framework’s most consequential structural fact — the number of spatial directions — traces back to an operational commitment rather than a discovered law echoes an older operationalist tradition, that a concept’s meaning is exhausted by the operations used to measure it [32]; A5’s insistence that only ratios of scale are observable is the same commitment applied to the scale field itself. Whether physics should be built this way — grounding its most basic structural facts in commitments about observation rather than discoveries about a mind-independent world, or whether that concedes too much to what can currently be measured — is a further instance of the realism question raised above, now sharpened to the single axiom doing the most structural work in this paper.

None of these questions is answered by 650 machine-checked theorems, and none was meant to be; a proof assistant adjudicates validity, not the questions this section has raised. What it can do, and what this paper has tried to do throughout, is state plainly which parts of the argument are forced by the axioms, which are physical identifications layered on top, and which are open — so that the questions above are inherited by the reader in the same sharp form they were found in, rather than smoothed into an ending that sounds more settled than the argument is.

Appendix A: Note on the Supplemental Material

The Supplemental Material gives, in the same order as the sections above, a complete mathematical restatement of every one of the 650 machine-verified theorems underlying this paper — full formal statements and full proofs, organized to mirror the structure of this paper section by section, with each item individually identifiable for cross-reference. This paper does not depend on the Supplemental Material to be read; the Supplemental Material exists so that every claim made informally above can be checked in complete formal detail.

-
- [1] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, Gw250114: Testing hawking’s area law and the kerr nature of black holes, *Physical Review Letters* **135**, 10.1103/kw5g-d732 (2025).
 - [2] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, Gw241011 and gw241110: Exploring binary formation and fundamental physics with asymmetric, high-spin black hole coalescences, *The Astrophysical Journal Letters* **993**, L21 (2025).
 - [3] L. H. THOMAS, The Motion of the Spinning Electron, *Nature* **117**, 514 (1926).
 - [4] J. Terrell, Invisibility of the Lorentz Contraction, *Physical Review* **116**, 1041 (1959).
 - [5] R. Penrose, The apparent shape of a relativistically moving sphere, *Proceedings of the Cambridge Philosophical Society* **55**, 137 (1959).

- [6] H. Weyl, Gravitation und Elektrizität, Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften , 465 (1918).
- [7] N. Cartwright, *The Dappled World: A Study of the Boundaries of Science* (Cambridge University Press, Cambridge, 1999).
- [8] S. B. Lambert and C. Le Poncin-Lafitte, Improved determination of γ by VLBI, *Astronomy & Astrophysics* **529**, A70 (2011).
- [9] R. V. Pound and G. A. Rebka, Apparent weight of photons, *Phys. Rev. Lett.* **4**, 337 (1960).
- [10] R. V. Pound and J. L. Snider, Effect of Gravity on Gamma Radiation, *Physical Review* **140**, 788 (1965).
- [11] R. A. Hulse and J. H. Taylor, Discovery of a pulsar in a binary system., *The Astrophysical Journal Letters* **195**, L51 (1975).
- [12] J. M. Weisberg and Y. Huang, Relativistic measurements from timing the binary pulsar psr b1913+16, *The Astrophysical Journal* **829**, 55 (2016).
- [13] M. Kramer, I. H. Stairs, R. N. Manchester, N. Wex, A. T. Deller, W. A. Coles, M. Ali, M. Burgay, F. Camilo, I. Cognard, T. Damour, G. Desvignes, R. D. Ferdman, P. C. C. Freire, S. Grondin, L. Guillemot, G. B. Hobbs, G. Janssen, R. Karuppusamy, D. R. Lorimer, A. G. Lyne, J. W. McKee, M. McLaughlin, L. E. Münch, B. B. P. Perera, N. Pol, A. Possenti, J. Sarkissian, B. W. Stappers, and G. Theureau, Strong-field gravity tests with the double pulsar, *Phys. Rev. X* **11**, 041050 (2021).
- [14] J. Bailey, K. Borer, F. Combley, H. Drumm, F. J. M. Farley, J. H. Field, W. Flegel, P. M. Hattersley, F. Krienen, F. Lange, E. Picasso, and W. von Rüden, Measurements of relativistic time dilatation for positive and negative muons in a circular orbit, *Nature* **268**, 301 (1977).
- [15] S. Navas, C. Amsler, T. Gutsche, C. Hanhart, J. J. Hernández-Rey, C. Lourenço, A. Masoni, M. Mikhasenko, R. E. Mitchell, C. Patrignani, C. Schwanda, S. Spanier, G. Venanzoni, C. Z. Yuan, K. Agashe, G. Aielli, B. C. Allanach, J. Alvarez-Muñiz, M. Antonelli, E. C. Aschenauer, D. M. Asner, K. Assamagan, H. Baer, S. Banerjee, R. M. Barnett, L. Baudis, C. W. Bauer, J. J. Beatty, J. Beringer, A. Bettini, O. Biebel, K. M. Black, E. Blucher, R. Bonventre, R. A. Briere, A. Buckley, V. D. Burkert, M. A. Bychkov, R. N. Cahn, Z. Cao, M. Carena, G. Casarosa, A. Ceccucci, A. Cerri, R. S. Chivukula, G. Cowan, K. Cranmer, V. Crede, O. Cremonesi, G. D'Ambrosio, T. Damour, D. de Florian, A. de Gouvêa, T. DeGrand, S. Demers, Z. Demiragli, B. A. Dobrescu, M. D'Onofrio, M. Doser, H. K. Dreiner, P. Eerola, U. Egede, S. Eidelman, A. X. El-Khadra, J. Ellis, S. C. Eno, J. Erler, V. V. Ezhela, A. Fava, W. Fetscher, B. D. Fields, A. Freitas, H. Gallagher, T. Gershon, Y. Gershtein, T. Gherghetta, M. C. Gonzalez-Garcia, M. Goodman, C. Grab, A. V. Gritsan, C. Grojean, D. E. Groom, M. Grünewald, A. Gurtu, H. E. Haber, M. Hamel, S. Hashimoto, Y. Hayato, A. Hebecker, S. Heinemeyer, K. Hikasa, J. Hisano, A. Höcker, J. Holder, L. Hsu, J. Huston, T. Hyodo, A. Ianni, M. Kado, M. Karliner, U. F. Katz, M. Kenzie, V. A. Khoze, S. R. Klein, F. Krauss, M. Kreps, P. Krizan, B. Krusche, Y. Kwon, O. Lahav, L. P. Lellouch, J. Lesgourgues, A. R. Liddle, Z. Ligeti, C.-J. Lin, C. Lippmann, T. M. Liss, A. Lister, L. Littenberg, K. S. Lugovsky, S. B. Lugovsky, A. Lusiani, Y. Makida, F. Maltoni, A. V. Manohar, W. J. Marciano, J. Matthews, U.-G. Meißner, I.-A. Melzer-Pellmann, P. Mertsch, D. J. Miller, D. Milstead, K. Mönig, P. Molaro, F. Moortgat, M. Moskvic, N. Nagata, K. Nakamura, M. Narain, P. Nason, A. Nelles, M. Neubert, Y. Nir, H. B. O'Connell, C. A. J. O'Hare, K. A. Olive, J. A. Peacock, E. Pianori, A. Pich, A. Piepke, F. Pietropaolo, A. Pomarol, S. Pordes, S. Profumo, A. Quadt, K. Rabbertz, J. Rademacker, G. Raffelt, M. Ramsey-Musolf, P. Richardson, A. Ringwald, D. J. Robinson, S. Roesler, S. Rolli, A. Romanziouk, L. J. Rosenberg, J. L. Rosner, G. Rybka, M. G. Ryskin, R. A. Ryutin, B. Safdi, Y. Sakai, S. Sarkar, F. Sauli, O. Schneider, S. Schönert, K. Scholberg, A. J. Schwartz, J. Schwiening, D. Scott, F. Sefkow, U. Seljak, V. Sharma, S. R. Sharpe, V. Shiltsev, G. Signorelli, M. Silari, F. Simon, T. Sjöstrand, P. Skands, T. Skwarnicki, G. F. Smoot, A. Soffer, M. S. Sozzi, C. Spiering, A. Stahl, Y. Sumino, F. Takahashi, M. Tanabashi, and J. Tanaka, Review of particle physics*, *Physical Review D* **110**, 030001 (2024).
- [16] C. J. Morningstar and M. Peardon, Glueball spectrum from an anisotropic lattice study, *Phys. Rev. D* **60**, 034509 (1999).
- [17] F. Bayen, M. Flato, C. Fronsdal, A. Lichnerowicz, and D. Sternheimer, Deformation theory and quantization. i. deformations of symplectic structures, *Annals of Physics* **111**, 61 (1978).
- [18] M. Abdul Karim, J. Aguilar, S. Ahlen, S. Alam, L. Allen, C. A. Prieto, O. Alves, A. Anand, U. Andrade, E. Armengaud, A. Aviles, S. Bailey, C. Baltay, P. Bansal, A. Bault, J. Behera, S. BenZvi, D. Bianchi, C. Blake, S. Brieden, A. Brodzeller, D. Brooks, E. Buckley-Geer, E. Burtin, R. Calderon, R. Canning, A. C. Rosell, P. Carrilho, L. Casas, F. Castander, M. Charles, E. Chaussidon, J. Chaves-Montero, D. Chebat, X. Chen, T. Claybaugh, S. Cole, A. Cooper, A. Cuceu, K. Dawson, A. de la Macorra, A. de Mattia, N. Deiosso, J. Della Costa, R. Demina, A. Dey, B. Dey, Z. Ding, P. Doel, J. Edelstein, D. Eisenstein, W. Elbers, P. Fagrelus, K. Fanning, E. Fernández-García, S. Ferraro, A. Font-Ribera, J. Forero-Romero, C. Frenk, C. Garcia-Quintero, L. Garrison, E. Gaztañaga, H. Gil-Marín, S. G. A. Gontcho, D. Gonzalez, A. Gonzalez-Morales, C. Gordon, D. Green, G. Gutierrez, J. Guy, B. Hadzhiyska, C. Hahn, S. He, M. Herbold, H. Herrera-Alcantar, M.-F. Ho, K. Honscheid, C. Howlett, D. Huterer, M. Ishak, S. Juneau, N. Kamble, N. Karaçaylı, R. Kehoe, S. Kent, A. Kim, D. Kirkby, T. Kisner, S. Koposov, A. Kremin, A. Krolewski, O. Lahav, C. Lamman, M. Landriau, D. Lang, J. Lasker, J. Le Goff, L. Le Guillou, A. Leauthaud, M. Levi, Q. Li, T. Li, K. Lodha, M. Lokken, F. Lozano-Rodríguez, C. Magneville, M. Manera, P. Martini, W. Matthews, A. Meisner, J. Mena-Fernández, A. Menegas, T. Mergulhão, R. Miquel, J. Moustakas, A. Muñoz-Gutiérrez, D. Muñoz-Santos, A. Myers, S. Nadathur, K. Naidoo, L. Napolitano, J. Newman, G. Niz, H. Noriega, E. Paillas, N. Palanque-Delabrouille, J. Pan, J. Peacock, M. Ibanez, W. Percival, A. Pérez-Fernández, I. Pérez-Ràfols, M. Pieri, C. Poppett, F. Prada, D. Rabinowitz, A. Raichoor, C. Ramírez-Pérez, M. Rashkovetskyi, C. Ravoux, J. Rich, A. Rocher, C. Rockosi, J. Rohlf, J. Román-Herrera, A. Ross, G. Rossi, R. Ruggeri, V. Ruhlmann-Kleider, L. Samushia, E. Sanchez, N. Sanders, D. Schlegel, M. Schubnell, H. Seo, A. Shafieloo, R. Sharples, J. Silber, F. Sinigaglia, D. Sprayberry, T. Tan, G. Tarlé, P. Taylor, W. Turner, L. Ureña-López, R. Vaisakh, F. Valdes, G. Valogiannis, M. Vargas-Magaña, L. Verde, M. Walther, B. Weaver, D. Weinberg, M. White, M. Wolfson, C. Yèche,

- J. Yu, E. Zaborowski, P. Zarrouk, Z. Zhai, H. Zhang, C. Zhao, G. Zhao, R. Zhou, and H. a. Zou, Desi dr2 results. ii. measurements of baryon acoustic oscillations and cosmological constraints, *Physical Review D* **112**, 10.1103/tr6y-kpc6 (2025).
- [19] A. A. Abdo, M. Ackermann, M. Ajello, K. Asano, W. B. Atwood, M. Axelsson, L. Baldini, J. Ballet, G. Barbiellini, M. G. Baring, D. Bastieri, K. Bechtol, R. Bellazzini, B. Berenji, P. N. Bhat, E. Bissaldi, E. D. Bloom, E. Bonamente, J. Bonnell, A. W. Borgland, A. Bouvier, J. Bregeon, A. Brez, M. S. Briggs, M. Brigida, P. Bruel, J. M. Burgess, T. H. Burnett, G. A. Caliendo, R. A. Cameron, P. A. Caraveo, J. M. Casandjian, C. Cecchi, Ö. Çelik, V. Chaplin, E. Charles, C. C. Cheung, J. Chiang, S. Ciprini, R. Claus, J. Cohen-Tanugi, L. R. Cominsky, V. Connaughton, J. Conrad, S. Cutini, C. D. Dermer, A. de Angelis, F. de Palma, S. W. Digel, B. L. Dingus, E. do Couto e Silva, P. S. Drell, R. Dubois, D. Dumora, C. Farnier, C. Favuzzi, S. J. Fegan, J. Finke, G. Fishman, W. B. Focke, L. Foschini, Y. Fukazawa, S. Funk, P. Fusco, F. Gargano, D. Gasparrini, N. Gehrels, S. Germani, L. Gibby, B. Giebels, N. Giglietto, F. Giordano, T. Glanzman, G. Godfrey, J. Granot, J. Greiner, I. A. Grenier, M.-H. Grondin, J. E. Grove, D. Grupe, L. Guillemot, S. Guiriec, Y. Hanabata, A. K. Harding, M. Hayashida, E. Hays, E. A. Hoversten, R. E. Hughes, G. Jóhannesson, A. S. Johnson, R. P. Johnson, W. N. Johnson, T. Kamae, H. Katagiri, J. Kataoka, N. Kawai, M. Kerr, R. M. Kippen, J. Knödseder, D. Kocevski, C. Kouveliotou, F. Kuehn, M. Kuss, J. Lande, L. Latronico, M. Lemoine-Goumard, F. Longo, F. Loparco, B. Lott, M. N. Lovellette, P. Lubrano, G. M. Madejski, A. Makeev, M. N. Mazziotta, S. McBreen, J. E. McEnery, S. McGlynn, P. Mészáros, C. Meurer, P. F. Michelson, W. Mitthumsiri, T. Mizuno, A. A. Moiseev, C. Monte, M. E. Monzani, E. Moretti, A. Morselli, I. V. Moskalenko, S. Murgia, T. Nakamori, P. L. Nolan, J. P. Norris, E. Nuss, M. Ohno, T. Ohsugi, N. Omodei, E. Orlando, J. F. Ormes, M. Ozaki, W. S. Paciesas, D. Paneque, J. H. Panetta, D. Parent, V. Pelassa, M. Pepe, M. Pesce-Rollins, V. Petrosian, F. Piron, T. A. Porter, R. Preece, S. Rainò, E. Ramirez-Ruiz, R. Rando, M. Razzano, S. Razzaque, A. Reimer, O. Reimer, T. Reposeur, S. Ritz, L. S. Rochester, A. Y. Rodriguez, M. Roth, F. Ryde, H. F.-W. Sadrozinski, D. Sanchez, A. Sander, P. M. Saz Parkinson, J. D. Scargle, T. L. Schalk, C. Sgrò, E. J. Siskind, D. A. Smith, P. D. Smith, G. Spandre, P. Spinelli, M. Stamatikos, F. W. Stecker, M. S. Strickman, D. J. Suson, H. Tajima, H. Takahashi, T. Takahashi, T. Tanaka, J. B. Thayer, J. G. Thayer, D. J. Thompson, L. Tibaldo, K. Toma, D. F. Torres, G. Tosti, E. Troja, Y. Uchiyama, T. Uehara, T. L. Usher, A. J. van der Horst, V. Vasileiou, N. Vilchez, V. Vitale, A. von Kienlin, A. P. Waite, P. Wang, C. Wilson-Hodge, B. L. Winer, K. S. Wood, X. F. Wu, R. Yamazaki, T. Ylinen, and M. Ziegler, A limit on the variation of the speed of light arising from quantum gravity effects, *Nature* **462**, 331 (2009).
- [20] B. Bertotti, L. Iess, and P. Tortora, A test of general relativity using radio links with the Cassini spacecraft, *Nature* **425**, 374 (2003).
- [21] N. and Aghanim, Y. Akrami, M. Ashdown, J. Aumont, C. Baccigalupi, M. Ballardini, A. J. Banday, R. B. Barreiro, N. Bartolo, S. Basak, R. Battye, K. Benabed, J.-P. Bernard, M. Bersanelli, P. Bielewicz, J. J. Bock, J. R. Bond, J. Borrill, F. R. Bouchet, F. Boulanger, M. Bucher, C. Burigana, R. C. Butler, E. Calabrese, J.-F. Cardoso, J. Carron, A. Challinor, H. C. Chiang, J. Chluba, L. P. L. Colombo, C. Combet, D. Contreras, B. P. Crill, F. Cuttaia, P. de Bernardis, G. de Zotti, J. Delabrouille, J.-M. Delouis, E. Di Valentino, J. M. Diego, O. Doré, M. Douspis, A. Ducout, X. Dupac, S. Dusini, G. Efstathiou, F. Elsner, T. A. Enßlin, H. K. Eriksen, Y. Fantaye, M. Farhang, J. Fergusson, R. Fernandez-Cobos, F. Finelli, F. Forastieri, M. Frailis, A. A. Fraisse, E. Franceschi, A. Frolov, S. Galeotta, S. Galli, K. Ganga, R. T. Génova-Santos, M. Gerbino, T. Ghosh, J. González-Nuevo, K. M. Górski, S. Gratton, A. Gruppuso, J. E. Gudmundsson, J. Hamann, W. Handley, F. K. Hansen, D. Herranz, S. R. Hildebrandt, E. Hivon, Z. Huang, A. H. Jaffe, W. C. Jones, A. Karakci, E. Keihänen, R. Keskitalo, K. Kiiveri, J. Kim, T. S. Kisner, L. Knox, N. Krachmalnicoff, M. Kunz, H. Kurki-Suonio, G. Lagache, J.-M. Lamarre, A. Lasenby, M. Lattanzi, C. R. Lawrence, M. Le Jeune, P. Lemos, J. Lesgourgues, F. Levrier, A. Lewis, M. Liguori, P. B. Lilje, M. Lilley, V. Lindholm, M. López-Cañiego, P. M. Lubin, Y.-Z. Ma, J. F. Macías-Pérez, G. Maggio, D. Maino, N. Mandolesi, A. Mangilli, A. Marcos-Caballero, M. Maris, P. G. Martin, M. Martinelli, E. Martínez-González, S. Matarrese, N. Mauri, J. D. McEwen, P. R. Meinhold, A. Melchiorri, A. Mennella, M. Migliaccio, M. Millea, S. Mitra, M.-A. Miville-Deschênes, D. Molinari, L. Montier, G. Morgante, A. Moss, P. Natoli, H. U. Nørgaard-Nielsen, L. Pagano, D. Paoletti, B. Partridge, G. Patanchon, H. V. Peiris, F. Perrotta, V. Pettorino, F. Piacentini, L. Polastri, G. Polenta, J.-L. Puget, J. P. Rachen, M. Reinecke, M. Remazeilles, A. Renzi, G. Rocha, C. Rosset, G. Roudier, J. A. Rubiño-Martín, B. Ruiz-Granados, L. Salvati, M. Sandri, M. Savelainen, D. Scott, E. P. S. Shellard, C. Sirignano, G. Sirri, L. D. Spencer, R. Sunyaev, A.-S. Suur-Uski, J. A. Tauber, D. Tavagnacco, M. Tenti, L. Toffolatti, M. Tomasi, T. Trombetti, L. Valenziano, J. Valiviita, B. Van Tent, L. Vibert, P. Vielva, F. Villa, N. Vittorio, B. D. Wandelt, I. K. Wehus, M. White, S. D. M. White, A. Zacchei, and A. Zonca, $\mu\text{Planck}/\text{i}\mu\text{ 2018 results: Vi. cosmological parameters}$, *Astronomy & Astrophysics* **641**, A6 (2020).
- [22] I. Caprini, G. Colangelo, and H. Leutwyler, Mass and width of the lowest resonance in qcd, *Phys. Rev. Lett.* **96**, 132001 (2006).
- [23] R. García-Martín, R. Kamiński, J. R. Peláez, and J. R. de Elvira, Precise determination of the $f_0(600)$ and $f_0(980)$ pole parameters from a dispersive data analysis, *Phys. Rev. Lett.* **107**, 072001 (2011).
- [24] S. Abe, S. Asami, M. Eizuka, S. Futagi, A. Gando, Y. Gando, T. Gima, A. Goto, T. Hachiya, K. Hata, S. Hayashida, K. Hosokawa, K. Ichimura, S. Ieki, H. Ikeda, K. Inoue, K. Ishidoshiro, Y. Kamei, N. Kawada, Y. Kishimoto, M. Koga, M. Kurasawa, N. Maemura, T. Mitsui, H. Miyake, T. Nakahata, K. Nakamura, K. Nakamura, R. Nakamura, H. Ozaki, T. Sakai, H. Sambonsugi, I. Shimizu, J. Shirai, K. Shiraishi, A. Suzuki, Y. Suzuki, A. Takeuchi, K. Tamae, K. Ueshima, H. Watanabe, Y. Yoshida, S. Obara, A. Ichikawa, D. Chernyak, A. Kozlov, K. Nakamura, S. Yoshida, Y. Takemoto, S. Umehara, K. Fushimi, K. Kotera, Y. Urano, B. Berger, B. Fujikawa, J. Learned, J. Maricic, S. Axani, J. Smolksy, Z. Fu, L. Winslow, Y. Efremenko, H. Karwowski, D. Markoff, W. Tornow, S. Dell’Oro, T. O’Donnell, J. Detwiler, S. Enomoto, M. Decowski, C. Grant, A. Li, and H. a. Song, Search for the majorana nature of neutrinos in the inverted mass ordering region with kamland-zen, *Physical Review Letters* **130**, 10.1103/physrevlett.130.051801 (2023).

- [25] J. Earman, *World Enough and Space-Time: Absolute versus Relational Theories of Space and Time* (MIT Press, Cambridge, MA, 1989).
- [26] K. R. Popper, *The Logic of Scientific Discovery* (Hutchinson & Co., London, 1959) english translation; German original *Logik der Forschung*, 1934.
- [27] X. Yang, The philosophy crisis of theoretical physics (2026).
- [28] K. Gödel, Über formal unentscheidbare sätze der Principia Mathematica und verwandter Systeme I, *Monatshefte für Mathematik und Physik* **38**, 173 (1931).
- [29] S. Hawking, Gödel and the end of physics (2002), dirac Centennial Lecture, Cambridge.
- [30] F. Dyson, Birds and frogs, *Notices of the American Mathematical Society* **56**, 212 (2009).
- [31] J. D. Barrow, Gödel and physics, in *Kurt Gödel and the Foundations of Mathematics: Horizons of Truth*, edited by M. Baaz, C. H. Papadimitriou, H. Putnam, D. S. Scott, and C. L. Harper (Cambridge University Press, Cambridge, 2011) pp. 255–276.
- [32] P. W. Bridgman, *The Logic of Modern Physics* (Macmillan, New York, 1927).